

Online Appendix for: From Speech to Networks: Modeling Discursive Alignment in the U.S. Congress

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A The (T)ERGM Framework

The formal definition of an ERGM is

$$P(N = N^* | \boldsymbol{\theta}, \mathcal{N}) = \frac{\exp\{\boldsymbol{\theta}^\top \mathbf{h}(N)\}}{\kappa(\boldsymbol{\theta}, \mathcal{N})}, \quad \forall N^* \in \mathcal{N}, \quad (1)$$

where N represents a network and is a random variable within the network space $\mathcal{N}$. Here, $\mathbf{h}(N)$ denotes a vector of sufficient statistics calculated based on N , and $\boldsymbol{\theta}$ is a vector of parameters associated with these statistics. The term $\kappa(\boldsymbol{\theta}, \mathcal{N}) = \sum_{N \in \mathcal{N}} \exp\{\boldsymbol{\theta}^\top \mathbf{h}(N)\}$ is a normalizing constant, which represents the sum of the numerator over all possible networks (Hunter et al., 2008).

As probabilistic models for networks, ERGMs allow researchers to estimate the probability of observing specific networks, based on both endogenous network characteristics and exogenous covariates, in a manner conceptually similar to traditional regressions (Box-Steffensmeier et al., 2018; Shumate and Palazzolo, 2010; Cranmer and Desmarais, 2011; Cranmer et al., 2017; Desmarais and Cranmer, 2012). Within this framework, the observed network is viewed as one realization among the set of all possible networks that could be generated by an underlying data-generating process. However, unlike traditional models, ERGMs do not assume independence among observations. This flexibility allows for the modeling of endogenous dependencies within the network alongside exogenous covariate effects (Cranmer et al., 2020), distinguishing them from classical regression approaches. By accounting for the interdependence of edges, ERGMs provide a more realistic representation of social networks (Robins et al., 2007).

The formal definition of a TERGM is

$$P(N^t | N^{t-K}, \dots, N^{t-1}, \boldsymbol{\theta}) = \frac{\exp\{\boldsymbol{\theta}^\top \mathbf{h}(N^t, N^{t-1}, \dots, N^{t-K})\}}{\kappa(\boldsymbol{\theta}, N^{t-K}, \dots, N^{t-1})}. \quad (2)$$

The joint probability of observing multiple networks between times $K + 1$ and T is modeled by taking the product of the probabilities of the individual networks, conditional on the preceding networks. This can be expressed as

$$P(N^{K+1}, \dots, N^T | N^1, \dots, N^K, \boldsymbol{\theta}) = \prod_{t=K+1}^T P(N^t | N^{t-K}, \dots, N^{t-1}, \boldsymbol{\theta}). \quad (3)$$

A.1 Specification and estimation details

Our estimation sample comprises the eleven transitions between adjacent Congresses (104th–105th through 114th–115th), where each Congress’s network is modeled conditional on the previous one and restricted to members serving in both Congresses. The previous Congress’s tie is included in the model as an edge covariate (“lagged tie”). Restricting each transition to continuing members models tie formation among legislators present in both Congresses; Section A.5 reports an alternative node set that retains every member serving at any point between the 104th and 115th Congresses; this alternative specification yields substantively equivalent estimates. Table 1 details our full model specification, categorizing terms into

Table 1: Full BTERGM Model Specification

Category	Network Statistic	(t)ergm Term
Endogenous	Edge Count (Intercept)	edges
	Reciprocity (Mutual ties)	mutual
	Triadic Closure (GWESP, $\phi = 0.4$)	gwesp(0.4, type = "OTP")
	Open Triads (GWNSP, $\phi = 0.3$)	gwnsp(0.3, type = "OTP")
	Popularity (GW In-degree, $\phi = 0.3$)	gwidegree(0.3)
	Sociality (GW Out-degree, $\phi = 0.5$)	gwodegree(0.5)
Exogenous	<i>Temporal/Structural</i>	
	Tie Persistence ($t - 1$)	edgescov(lag_use)
	Linear Time Trend	timecov()
	<i>Partisan/Ideological</i>	
	Party Homophily	edgescov(party_use)
	Party Homophily (Time Interaction)	timecov(party_use)
	Ideology Distance (DW-NOMINATE 1)	absdiff(abs_nominate)
	Ideology Distance (CF score)	absdiff(recipient.cfscore)
	<i>Personal/Legislative</i>	
	Speech Volume (Popularity)	nodeicov(log_volume)
	Speech Volume (Sociality)	nodeicov(log_volume)
	Seniority Distance	absdiff(seniority)
	Urbanization Distance	absdiff(urban_perc)
	Committee Homophily	edgescov(comm_use)
	Gender Homophily	nodematch(female)
	Race Homophily (Black)	nodematch(black)
	Ethnicity Homophily (Latino)	nodematch(latino)
	Majority Leadership Homophily	nodematch(maj_leader)
	Minority Leadership Homophily	nodematch(min_leader)
	Committee Chair Homophily	nodematch(chair)
	Majority Leadership (Popularity)	nodeicov(maj_leader)
	Minority Leadership (Popularity)	nodeicov(min_leader)
	Committee Chair (Popularity)	nodeicov(chair)

Note: Decay parameters (ϕ) were fixed following pilot model goodness-of-fit assessments (see Section A.2) using the `fixed = TRUE` argument, which is omitted from `ergm` and `tergm` package terms above for brevity.

endogenous structural terms and exogenous nodal or dyadic covariates. The former account for structural network properties, whereas the latter control for external attributes of both individual (nodes) or pairs of (dyads) legislators and the districts they represent.

We specify geometrically weighted structural terms to capture higher-order endogenous dependencies, such as triadic closure and popularity (Snijders et al., 2006). These terms

apply a geometrically decreasing weight to the distribution of degrees and shared partners, which reduces the risk of model degeneracy and captures the diminishing marginal returns of network structural features (what [Hunter \(2007\)](#) refers to as “a sort of anti-preferential attachment” effect). We treat the decay parameters as fixed rather than estimating them. Doing so mitigates the instability and near-degeneracy frequently encountered when optimizing the parameter space of curved exponential families ([Schweinberger, 2011](#)), and renders the resulting coefficients directly interpretable as the change in the log-odds of tie formation ([Hunter, 2007](#)). Fixed decay parameter values were selected by goodness of fit on a coordinate-wise grid of pilot fits (see [Section A.2](#), which reports the grid and sensitivity of the estimates to the decay configuration). Exogenous terms are as described in the main text, including each member’s logged one-minute word volume as popularity and sociality covariates, a linear time trend, and the same-party-by-time interaction. Models are estimated by bootstrapped maximum pseudolikelihood ([Leifeld et al., 2018](#)) with $R = 1000$ replications unless noted otherwise.¹ All models are fit in R 4.5.3 with `btergm` 1.11.1 and `ergm` 4.12.0 on an Apple M5 computer with 24 GB of memory. Six CPU cores are used for parallel computing, along with a fixed random seed. Point estimates are deterministic, and the bootstrap intervals are reproducible given the same seed and number of cores.

[Table 2](#) reports the full model from the main text, alongside endogenous term-only and exogenous term-only models for comparison. [Figure 1](#) visualizes the full model coefficients as odds ratios. [Table 7](#) adds two robustness columns discussed below. The geometrically weighted out-degree term is estimated as a goodness-of-fit control for the out-degree distribution, as its inclusion improves the pooled miss rate from 0.367 to 0.325 (see [Section A.2](#)). However, this coefficient is not interpreted throughout this appendix. Under maximum pseudolikelihood estimation, the coefficient is on the scale of the term’s change statistic. Following [Hunter \(2007\)](#) and [Morris et al. \(2008\)](#), the change statistic for an outgoing tie added to a member with out-degree k , given decay parameter $\phi = 0.5$, is defined as:

$$(1 - \exp(-0.5))^k. \tag{4}$$

The row-wise thresholding rule fixes each network’s mean out-degree at its density, 4.5–7.3 percent of the possible ties, so k averages between 17.6 and 24.4 and the change statistic is of order 10^{-8} to 10^{-10} . The estimated coefficient is correspondingly large and its magnitude is not comparable across networks. Throughout this appendix, we mark coefficients with an asterisk when the 95% bootstrap confidence interval excludes zero.

¹The out-of-sample training model, the equal-length and alternative-threshold refits, the decay selection fits for the main model, the volume-form comparison, the decay parameter sensitivity fits for the co-sponsorship model, the scaling probe fits, and the ideology leave-one-out fits use $R = 100$, as noted where each is reported.

Table 2: **Bootstrapped TERGM estimates, one-minute speech networks, 104th-115th Congresses.** 95% bootstrap confidence intervals are displayed in parentheses, where * marks intervals that exclude zero.

	Full	Endogenous	Exogenous
Edges	-7.349 (-7.946;-6.715)*	-2.962 (-3.119;-2.779)*	-11.350 (-12.248;-10.496)*
Reciprocity	3.261 (3.034;3.477)*	3.074 (2.910;3.262)*	
Triadic Closure	1.112 (1.026;1.184)*	1.254 (1.152;1.345)*	
Open Triads	-0.232 (-0.250;-0.217)*	-0.280 (-0.298;-0.263)*	
Tie Persistence ($t - 1$)	0.485 (0.418;0.546)*	0.863 (0.810;0.920)*	
Linear Time Trend	-0.032 (-0.087;-0.001)*		-0.100 (-0.198;-0.052)*
Party Hom.	-0.007 (-0.283;0.203)		0.224 (-0.152;0.473)
Party Hom. $\times$ Time	0.019 (0.004;0.049)*		0.060 (0.036;0.112)*
DW-NOM 1 (Dist.)	-0.320 (-0.443;-0.185)*		-0.781 (-0.940;-0.602)*
CF Score (Dist.)	-0.213 (-0.273;-0.141)*		-0.144 (-0.251;-0.042)*
Speech Vol. (Pop.)	0.578 (0.531;0.629)*		1.234 (1.180;1.304)*
Speech Vol. (Soc.)	0.032 (0.008;0.063)*		0.013 (-0.030;0.061)
Seniority (Dist.)	-0.010 (-0.013;-0.006)*		-0.025 (-0.032;-0.017)*
Urban. (Dist.)	-0.192 (-0.330;-0.056)*		-0.427 (-0.602;-0.256)*
Comm. Hom.	0.069 (0.043;0.097)*		0.091 (0.053;0.137)*
Gender Hom.	-0.014 (-0.052;0.019)		-0.025 (-0.090;0.075)
Race Hom. (Black)	-0.013 (-0.075;0.062)		-0.081 (-0.185;0.028)
Eth. Hom. (Latino)	0.046 (-0.040;0.148)		0.058 (-0.046;0.188)
Maj. Leader Hom.	-0.038 (-0.102;0.021)		0.017 (-0.059;0.096)
Min. Leader Hom.	0.036 (-0.015;0.090)		-0.008 (-0.057;0.050)
Comm. Chair Hom.	0.034 (-0.024;0.092)		-0.013 (-0.089;0.057)
Maj. Leader (Pop.)	0.103 (-0.060;0.284)		0.437 (0.192;0.638)*
Min. Leader (Pop.)	0.101 (-0.116;0.357)		0.161 (-0.056;0.413)
Comm. Chair (Pop.)	0.136 (-0.037;0.271)		-0.266 (-0.583;0.035)

* Null hypothesis value outside the confidence interval.

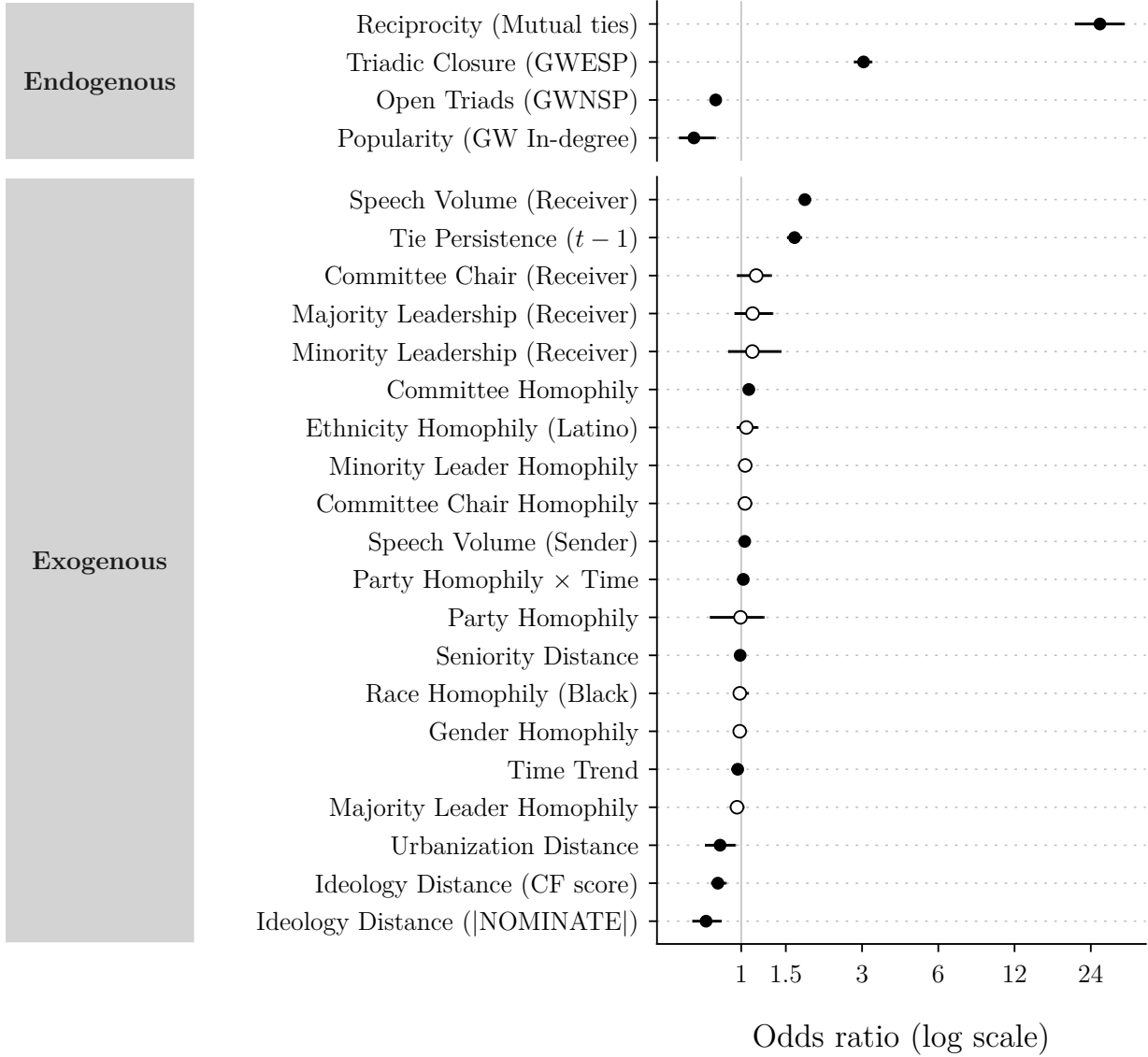


Figure 1: **Bootstrapped TERGM estimates, one-minute speech networks, 104th–115th Congresses.** Odds ratios (exponentiated estimates) for the full model of Table 2, plotted on a log scale so that distances correspond to log-odds; horizontal bars are 95% bootstrap confidence intervals ($R = 1000$), and the vertical line at 1 marks no effect. Filled points indicate intervals that exclude 1; open points denote intervals that include 1. Terms are ordered by magnitude within each group. The edges (intercept) and geometrically weighted out-degree terms are omitted; the latter’s magnitude reflects the scaling of its change statistic rather than a substantive effect (Section A.1).

A.2 Decay parameter selection and sensitivity

To select the decay parameters (denoted ϕ) of the geometrically weighted terms we evaluated fourteen candidate specifications across a coordinate-wise grid. Each pilot specification was estimated using $R = 100$ bootstrap replications, with goodness of fit evaluated over 30 simulated networks per temporal transition. Table 3 reports each configuration’s miss rate for the in-degree, out-degree, edgewise, and dyadwise shared partner distributions, together

with the mean (pooled) miss rate across the four statistics and the in-sample area under the receiver operating characteristic curve (AUC-ROC). For each bin of a distribution, the eleven observed counts are compared against the pooled simulation counts with a two-sample t -test. The miss rate is the proportion of bins in which that test rejects at the $\alpha = 0.05$ significance level.²

The selected configuration (GWESP = 0.4, GW-in-degree = 0.3, GWNSP = 0.3, and GW-out-degree = 0.5) achieves the lowest pooled miss rate at 0.325, against a next-best of 0.339. Notably, coordinate-wise optimal values did not stack additively; the optimal decay for one structural parameter depended heavily on the specification of the others. Predictive performance remained essentially flat across all configurations, with AUC-ROC scores tightly bounded between 0.897 and 0.904. This suggests that while precise decay tuning is necessary for accurately capturing the network’s higher-order degree and shared partner distributions, it is not necessary for baseline tie prediction.

Decays (ESP/ideg/NSP/odeg)	In-deg.	Out-deg.	ESP	DSP	Pooled	AUC-ROC
0.4 / 0.3 / 0.3 / 0.5 [†]	0.210	0.405	0.257	0.429	0.325	0.902
0.4 / 0.5 / 0.1 / –	0.210	0.455	0.308	0.385	0.339	0.900
0.7 / 0.3 / 0.3 / 0.5	0.206	0.405	0.289	0.474	0.343	0.904
0.7 / 0.3 / 0.1 / –	0.231	0.439	0.257	0.457	0.346	0.902
0.4 / 0.8 / 0.1 / –	0.189	0.512	0.263	0.436	0.350	0.900
0.7 / 0.5 / 0.3 / 0.5	0.244	0.463	0.250	0.444	0.350	0.903
0.2 / 0.3 / 0.1 / –	0.206	0.550	0.250	0.405	0.353	0.897
0.4 / 0.3 / 0.5 / –	0.223	0.550	0.270	0.405	0.362	0.903
0.4 / 0.3 / 0.05 / –	0.231	0.488	0.316	0.421	0.364	0.899
0.4 / 0.3 / 0.3 / –	0.210	0.548	0.263	0.447	0.367	0.902
0.4 / 0.3 / 0.1 / –	0.239	0.537	0.306	0.444	0.382	0.899
0.4 / 0.1 / 0.1 / –	0.176	0.535	0.342	0.474	0.382	0.898
0.7 / 0.5 / 0.3 / –	0.227	0.595	0.270	0.486	0.395	0.904
1 / 0.3 / 0.1 / –	0.256	0.581	0.444	0.474	0.439	0.902

Table 3: **Decay-selection grid.** Goodness-of-fit miss rates (lower is better) by statistic for the fourteen candidate decay configurations, ordered by pooled miss rate. Configurations are labeled by their GWESP / GW-in-degree / GWNSP / GW-out-degree decays (“–” = no out-degree term); [†] marks the configuration used in the final specification. Pilot models were fit using $R = 100$ bootstrap replications with 30 goodness-of-fit simulations per transition.

Table 3 reports the goodness of fit miss rates by statistic for the fourteen candidate decay parameter configurations. Close examination of the rates across the grid reveals that the contribution of the geometrically weighted out-degree term is isolated. Looking specifically at the chosen configuration, removing this term while leaving the final configuration otherwise unchanged (fifth row from the bottom), the out-degree miss rate worsens from

²Because the test compares the observed counts against the pooled simulated draws, its power (and with it the miss rate) increases with the number of simulations. Miss rates are therefore comparable only across models evaluated with the same number of simulated networks, which is why the magnitude-based measure reported in Section B is used for comparison across fits.

0.405—notably tied for the best among all grid configurations—to 0.548. Similarly, the edgewise and dyadwise shared partner statistics’ miss rates both worsen after the out-degree term is excluded. We rely on the pooled criterion because the degree, edgewise, and dyadwise distributions matter jointly. To verify that our substantive findings are robust to this choice, Table 4 reports the sensitivity of the coefficient estimates to the decay configuration, including the minimum, maximum, and range of each covariate estimate across the fourteen pilot fits.

Term	Minimum	Maximum	Range
Sociality	111.170	114.054	2.884
Edges	−7.464	−5.880	1.584
Reciprocity	2.549	3.396	0.848
Triadic Closure	0.814	1.136	0.322
Popularity	−0.550	−0.245	0.306
Speech Vol. (Pop.)	0.452	0.635	0.183
Speech Vol. (Soc.)	−0.113	0.060	0.173
DW-NOM 1 (Dist.)	−0.338	−0.253	0.086
Tie Persistence ($t - 1$)	0.456	0.519	0.063
Maj. Leader (Pop.)	0.075	0.126	0.051
Urbanization (Dist.)	−0.204	−0.163	0.041
Party Hom.	−0.036	0.004	0.039
Race Hom. (Black)	−0.016	0.019	0.035
Comm. Chair (Pop.)	0.121	0.156	0.035
Maj. Leader Hom.	−0.051	−0.019	0.031
CF Score (Dist.)	−0.237	−0.206	0.030
Min. Leader (Pop.)	0.081	0.111	0.030
Open Triads	−0.243	−0.217	0.026
Gender Hom.	−0.018	0.008	0.026
Min. Leader Hom.	0.027	0.048	0.021
Comm. Chair Hom.	0.028	0.045	0.017
Eth. Hom. (Latino)	0.046	0.061	0.015
Comm. Hom.	0.068	0.082	0.014
Linear Time Trend	−0.037	−0.025	0.012
Party Hom. $\times$ Time	0.014	0.022	0.008
Seniority (Dist.)	−0.010	−0.009	0.002

Table 4: **Coefficient sensitivity to the decay configuration.** Minimum, maximum, and range of each estimate across the fourteen pilot configurations of Table 3, ordered by range. Estimates from $R = 100$ pilot fits. The geometrically weighted terms are pooled across their own decay values, so their ranges—and that of the edges term, the intercept against which they re-level—partly reflect the changing decay parameterization rather than instability (the GW out-degree row additionally inherits that term’s large change-statistic scaling).

Turning to Table 4’s sensitivity results, the exogenous covariates are largely insensitive to the choice of decay parameters. The largest ranges are the two volume terms (0.18 for popularity and 0.17 for sociality), followed by the |NOMINATE| difference term at 0.09,

with sixteen of the twenty-six terms falling below 0.05. For most covariates that range is smaller than the corresponding confidence interval—the |NOMINATE| difference spans 0.09 against an interval width 0.26, the CF-score difference 0.03 against 0.13, and urbanization 0.04 against 0.27—but the two volume terms are exceptions, with each spanning more than its own interval (0.10 and 0.06, respectively). Accordingly, the sign of statistically significant coefficients is consistent with the final model across the grid for all but one covariate, the speech volume sociality (sender) term. On the other hand, the endogenous terms are more variable. The reciprocity coefficient ranges over 0.85 log-odds, triadic closure over 0.32, and popularity over 0.31, while the edges (intercept) term ranges over 1.58. The temporal terms, by contrast, are tighter, with the lagged-tie term ranging over 0.06 and the party-by-time interaction over 0.008. This higher variability is expected, as the geometrically weighted statistics partition overlapping network structure and re-weighting one re-apportions shared variance among the others. This phenomenon in part motivates fixing the decay parameters to goodness of fit-selected values rather than treating them as estimable. The structural coefficients are only interpretable jointly with their decay configuration, whereas the exogenous covariate estimates—which correspond to the inferential hypotheses on ideology, committee, urbanization, seniority, volume, and partisan dynamics—do not depend on it. Reciprocity is positive and significant in all fourteen configurations, which, as Section H shows, reflects the construction of the networks rather than mutual alignment.

A.3 Raw-NOMINATE ideology term

The main specification measures ideology with the absolute difference in members’ *absolute* DW-NOMINATE scores (ideological intensity) and the absolute difference in CF scores. We exclude the absolute difference in *raw* DW-NOMINATE scores from the main model because it is nearly collinear with the CF-score difference across the 745,032 pooled dyads ($r = 0.90$). When the raw difference is included along with CF scores (Table 7, Column 2), the model cannot reliably disentangle the competing signals. Significance shifts to the raw term (-0.220 , 95% CI $[-0.417, -0.035]$), the intensity term’s interval widens to include zero and its magnitude shrinks (-0.202 , 95% CI $[-0.398, 0.027]$, compared to -0.32^* in the main model), and the CF-score coefficient decreases in magnitude from -0.213 to -0.164 while remaining significant. Because which of the three collinear distances carries the significance is not informative about the underlying structure, our extremist-versus-centrist interpretation relies exclusively on the main model. In contrast, the correlation between the two retained metrics—intensity and CF-score differences—is low ($r = 0.08$). These two ideology terms do not interfere with one another, as re-estimating the model with each term independently gives similar estimates (the intensity estimate is -0.32 when modeled jointly with CF score versus -0.38 alone; CF score is -0.21 when modeled jointly versus -0.23 alone), with neither term’s significance depending on the other.

A.4 Quadratic party-by-time term

The descriptive party-assortativity series is non-monotone, declining to a trough in the 107th Congress before rising. To test whether the conditional partisan trend shares this curvature, we re-estimated the full specification by adding a centered quadratic time transform, $(t - 6)^2$

over the eleven transitions, as both a main effect and an interaction with the same-party indicator ($R = 1000$). Centering the quadratic term at the midpoint of the transition index leaves it uncorrelated with the linear time term, so the two do not compete for the same variation. Neither the quadratic interaction term (0.001, 95% CI $[-0.016, 0.017]$) nor the quadratic main effect (0.001, $[-0.027, 0.033]$) is statistically significant (Table 5), and the linear party-by-time point estimate is unchanged by their addition to the model (0.019, $[0.004, 0.070]$), though its interval widens. No other interpreted coefficient moves by more than 0.011 in log-odds. We therefore find no evidence that the linear interaction masks an underlying conditional curvature, though a null finding does not rule out modest curvature within the confidence interval.

	Main model	+ Quadratic time
Party hom.	-0.007 $[-0.283, 0.203]$	-0.017 $[-0.436, 0.207]$
Party hom. $\times$ time	0.019* $[0.004, 0.049]$	0.019* $[0.004, 0.070]$
Party hom. $\times (t - 6)^2$		0.001 $[-0.016, 0.017]$
Time trend	-0.032* $[-0.087, -0.001]$	-0.032* $[-0.133, -0.003]$
$(t - 6)^2$		0.001 $[-0.027, 0.033]$

Table 5: **Quadratic party-by-time term.** Party and time terms of the main model and of the same specification with centered quadratic time terms, $(t - 6)^2$, added as a main effect and as an interaction with the same-party indicator ($R = 1000$ for both). 95% bootstrap confidence intervals in brackets; * indicates the interval excludes zero.

A.5 Composition-change robustness

Congressional membership fluctuates from Congress to Congress, creating a dynamic node set. We address this variation with two approaches. Our main text models employ an intersection approach, restricting each transition to members serving in both adjacent Congresses. While this effectively analyzes tie formation among continuing members, it necessitates the exclusion of freshmen and departing members. As shown in Table 6, 66 to 80 percent (73% on average) of each Congress’s members in our dataset also gave a speech in the previous Congress and are included in our main text model. The remaining members (73 to 129 per Congress)—who are either newly elected or gave no one-minute speeches in the previous Congress—are excluded. The number of dyads is proportional to the number of members squared, and so the intersection approach retains a smaller share of the ordered pairs (44 to 64 percent) and of the observed ties (39 to 66 percent). To assess the robustness of this specification, we implement an all-member expansion, or a “union” set, retaining every representative present at any point during the sample period. Each network is embedded within a common frame of 1,118 members, with non-serving legislators represented as isolated nodes. This approach yields a stable node set, which facilitates the use of an autoregressive memory term to anchor alignment persistence for the entire cohort. However, it creates a high proportion of structurally impossible dyads—roughly ninety percent per network—where a tie is precluded by the absence of at least one member, irrespective of the influence of any covariate or network process.

The union node set creates structural zeros (dyads that are structurally impossible), which are distinct from observed non-ties where a relationship simply did not form. Treating these structural zeros as ordinary non-ties is inaccurate; under such an approach, the model would evaluate the likelihood as if impossible ties were in fact possible but simply absent. This mismatch induces bias in the edge-count parameter and other structural statistic estimates by forcing them to be estimated against an unrealistic baseline. The standard methodological remedy is an offset, which fixes a model coefficient at a specific value rather than estimating it, thereby adjusting comparisons across networks with varying sizes and compositions (Krivitsky et al., 2011).

In the ERGM framework, the probability of an edge between two nodes, $P(Y_{ij} = 1)$, is modeled via a logit link function that relates the probability to the linear predictor:

$$\text{logit}(P(Y_{ij} = 1)) = \log \left(\frac{P(Y_{ij} = 1)}{1 - P(Y_{ij} = 1)} \right), \quad (5)$$

where $\log$ denotes the natural logarithm. For dyads that are structurally impossible, the probability of a tie is strictly zero. To force this outcome in the model, we determine the required value of the linear predictor:

$$\lim_{P(Y_{ij}=1) \rightarrow 0^+} \log \left(\frac{P(Y_{ij} = 1)}{1 - P(Y_{ij} = 1)} \right) = -\infty. \quad (6)$$

Because the natural logarithm diverges to $-\infty$ as $P(Y_{ij} = 1)$ approaches zero from the right, the coefficient associated with an indicator variable for an impossible dyad is $-\infty$. This sets the probability of the tie to zero, effectively removing these dyads from the set of all dyads the model considers as potential tie locations and preventing the estimation of ties where they cannot exist.

While the `ergm` package natively supports fixed offsets, the bootstrapped MPLE estimator for our temporal model lacks this support. We implement an equivalent restriction by including the impossible-dyad indicator as an edge covariate. Under maximum pseudolikelihood, estimation is a logistic regression over dyads. Because the impossible-dyad indicator perfectly predicts non-ties (every dyad it indicates is observed as a zero), the model exhibits perfect separation. As a result, the maximizing coefficient is theoretically unbounded below. As this coefficient diverges, the fitted tie probability of every impossible dyad converges to zero and those dyads do not contribute to the score equations of the other parameters. In the limit, the remaining coefficients are estimated as if the impossible dyads were deleted, mirroring the behavior of a true offset. In practice, however, the optimizer stops at a large finite value where the likelihood is flat. Our model returns -268.16 , with a bootstrap confidence interval spanning $[-1005.09, -167.53]$. The width of that interval across resamples indicates that the coefficient is a stopping point on a flat surface rather than an estimated quantity to be interpreted.

We verify with two empirical checks that this construction behaves as a true offset and does not contaminate the estimates. First, we conduct a scaling probe; if the impossible dyads are excluded, then multiplying the indicator covariate by any factor should rescale its estimated coefficient by the reciprocal (preserving the linear predictor) while leaving the other estimates unchanged. Conducting this procedure with a scaling factor of ten, in a pilot fit

at $R = 100$ the coefficient moves from -15.759 to -1.5759 . Moreover, the maximum change in any substantive coefficient is only 1.1×10^{-11} . Next, we compare the union-set approach against the full model in the main text, which employs the intersection approach that deletes non-serving members by design. As shown in the third column of Table 7, the intersection and union approaches agree on the sign of 24 out of 25 shared terms (the exception is an insignificant Black-homophily term). They also agree on significance for 24 of 25, with the exception being the committee chair popularity term, which is insignificant in the main model and significant under the union set. Endogenous and speech volume terms display high correspondence, with reciprocity within 2%, edgewise shared partners within 6%, and speech volume popularity within approximately 5%. The dyadic ideology and urbanization terms agree within 16%. Although the two smallest-magnitude covariates shift more in relative terms (shared committee moves from 0.07 to 0.10, and seniority goes from -0.010 to -0.014), they maintain consistent signs and statistical significance across both specifications. The two specifications operationalize persistence differently and thus share 25 rather than 26 terms. The intersection model includes a lagged-tie covariate (0.485, 95% CI [0.418, 0.546]), while the fixed frame of the union set incorporates a first-order autoregressive memory term (0.398, 95% CI [0.333, 0.457]). Both are positive and significant with comparable magnitude, indicating that alignment persistence is not an artifact of how the node set is handled to facilitate temporal modeling.

Transition	n_t	Continuing	Entering	Members kept	Dyads kept	Ties kept
104–105	352	263	89	0.75	0.56	0.52
105–106	335	260	75	0.78	0.60	0.66
106–107	322	243	79	0.75	0.57	0.58
107–108	312	216	96	0.69	0.48	0.53
108–109	314	235	79	0.75	0.56	0.61
109–110	335	227	108	0.68	0.46	0.39
110–111	387	273	114	0.71	0.50	0.55
111–112	349	237	112	0.68	0.46	0.53
112–113	385	256	129	0.66	0.44	0.42
113–114	371	298	73	0.80	0.64	0.64
114–115	374	282	92	0.75	0.57	0.55
Mean	349	254	95	0.73	0.53	0.55

Table 6: **Members and ties kept by the continuing-member restriction.** For each transition, n_t is the number of speaking members in Congress t . Continuing members also spoke in Congress $t - 1$ and enter the estimation; entering members did not. The last three columns give the shares of members, ordered pairs, and observed ties in Congress t that the restriction retains.

Table 7: **Robustness: raw-NOMINATE ideology term and the union-node-set composition.** The first column repeats the full model of Table 2. 95% bootstrap confidence intervals in parentheses.

	Full	+ raw NOMINATE	Union node set (arch A)
Edges	-7.349 (-7.946;-6.715)*	-7.257 (-7.863;-6.618)*	-7.230 (-8.028;-6.478)*
Reciprocity	3.261 (3.034;3.477)*	3.261 (3.034;3.477)*	3.318 (3.090;3.544)*
Triadic Closure	1.112 (1.026;1.184)*	1.112 (1.026;1.184)*	1.173 (1.085;1.240)*
Open Triads	-0.232 (-0.250;-0.217)*	-0.232 (-0.250;-0.217)*	-0.211 (-0.227;-0.198)*
Tie Persistence ($t - 1$)	0.485 (0.418;0.546)*	0.485 (0.418;0.546)*	
Linear Time Trend	-0.032 (-0.087;-0.001)*	-0.032 (-0.087;-0.001)*	-0.033 (-0.088;-0.003)*
Party Hom.	-0.007 (-0.283;0.203)	-0.096 (-0.388;0.141)	-0.015 (-0.294;0.191)
Party Hom. $\times$ Time	0.019 (0.004;0.049)*	0.019 (0.004;0.049)*	0.017 (0.002;0.046)*
DW-NOM 1 (Dist.)	-0.320 (-0.443;-0.185)*	-0.202 (-0.398;0.027)	-0.270 (-0.392;-0.155)*
CF Score (Dist.)	-0.213 (-0.273;-0.141)*	-0.164 (-0.239;-0.093)*	-0.235 (-0.299;-0.174)*
DW-NOM 1 (Dist.)		-0.220 (-0.417;-0.035)*	
Speech Vol. (Pop.)	0.578 (0.531;0.629)*	0.580 (0.533;0.630)*	0.552 (0.494;0.619)*
Speech Vol. (Soc.)	0.032 (0.008;0.063)*	0.032 (0.008;0.064)*	0.050 (0.023;0.084)*
Seniority (Dist.)	-0.010 (-0.013;-0.006)*	-0.010 (-0.013;-0.006)*	-0.014 (-0.019;-0.009)*
Urban. (Dist.)	-0.192 (-0.330;-0.056)*	-0.187 (-0.322;-0.051)*	-0.204 (-0.275;-0.124)*
Comm. Hom.	0.069 (0.043;0.097)*	0.069 (0.043;0.097)*	0.098 (0.073;0.122)*
Gender Hom.	-0.014 (-0.052;0.019)	-0.012 (-0.051;0.020)	-0.021 (-0.045;0.005)
Race Hom. (Black)	-0.013 (-0.075;0.062)	-0.026 (-0.087;0.048)	0.031 (-0.021;0.090)
Eth. Hom. (Latino)	0.046 (-0.040;0.148)	0.049 (-0.038;0.149)	0.037 (-0.042;0.116)
Maj. Leader Hom.	-0.038 (-0.102;0.021)	-0.044 (-0.108;0.013)	-0.020 (-0.072;0.039)
Min. Leader Hom.	0.036 (-0.015;0.090)	0.034 (-0.020;0.088)	0.015 (-0.038;0.074)
Comm. Chair Hom.	0.034 (-0.024;0.092)	0.031 (-0.028;0.090)	0.057 (-0.002;0.109)
Maj. Leader (Pop.)	0.103 (-0.060;0.284)	0.103 (-0.062;0.283)	0.041 (-0.128;0.217)
Min. Leader (Pop.)	0.101 (-0.116;0.357)	0.099 (-0.121;0.358)	0.080 (-0.113;0.318)
Comm. Chair (Pop.)	0.136 (-0.037;0.271)	0.138 (-0.035;0.273)	0.206 (0.055;0.322)*
Str. Zero (Offset)			-268.164 (-1005.092;-167.530)*

* Null hypothesis value outside the confidence interval.

B Goodness of Fit

Figure 2 compares the observed networks with networks simulated from the fitted full model on the in-degree, out-degree, edgewise shared partner, and dyadwise shared partner distributions.³ One hundred networks are simulated per transition. We summarize fit by the aggregate relative misfit,

$$\frac{\sum_k |\text{obs}_k - \text{sim}_k|}{\sum_k |\text{obs}_k|},$$

over the distribution’s bins. This is a magnitude-based measure that, unlike significance-based miss rates, is comparable across simulation counts. The model reproduces the dyadwise shared partner distribution to within 15 percent and the in-degree distribution to within 25 percent, with only 11 percent of the observed in-degree mass falling in bins whose observed value differs significantly from the simulated distribution. The edgewise shared partner distribution fits less well, with aggregate misfit of 33 percent. The fitted model produces too many edges with roughly nine to fifteen shared partners and too few with three to six, capturing the aggregate level of transitivity but over-smoothing its distribution. This imperfection is present across the entire decay grid (Section A.2) and therefore a property of the geometrically weighted specification family rather than of the decay choice. Because the substantive estimates are stable across the full grid (Table 4), with the exception of the sender-volume term, this misfit largely affects how well the model simulates the shared partner profile. Out-degree exhibits the poorest fit out of every specification containing the endogenous terms, with aggregate misfit at 53 percent. This misfit is by construction, since row-wise thresholding at $z > 1.64$ holds each member’s out-degree to a narrow band around 4.5 to 7.3 percent of the network size. Such z -score thresholding leaves little variance for any model to fit. Pooled across members, its coefficient of variation is 0.20, against 1.90 for in-degree. We therefore caution against interpreting out-degree fit as a discriminating statistic for these networks.

³The pooled geodesic-distance statistic is unavailable for this model. `btergm`’s implementation cannot pool geodesic tables across networks of varying size. Degree and shared partner (edgewise and dyadwise) fit is reported in full.

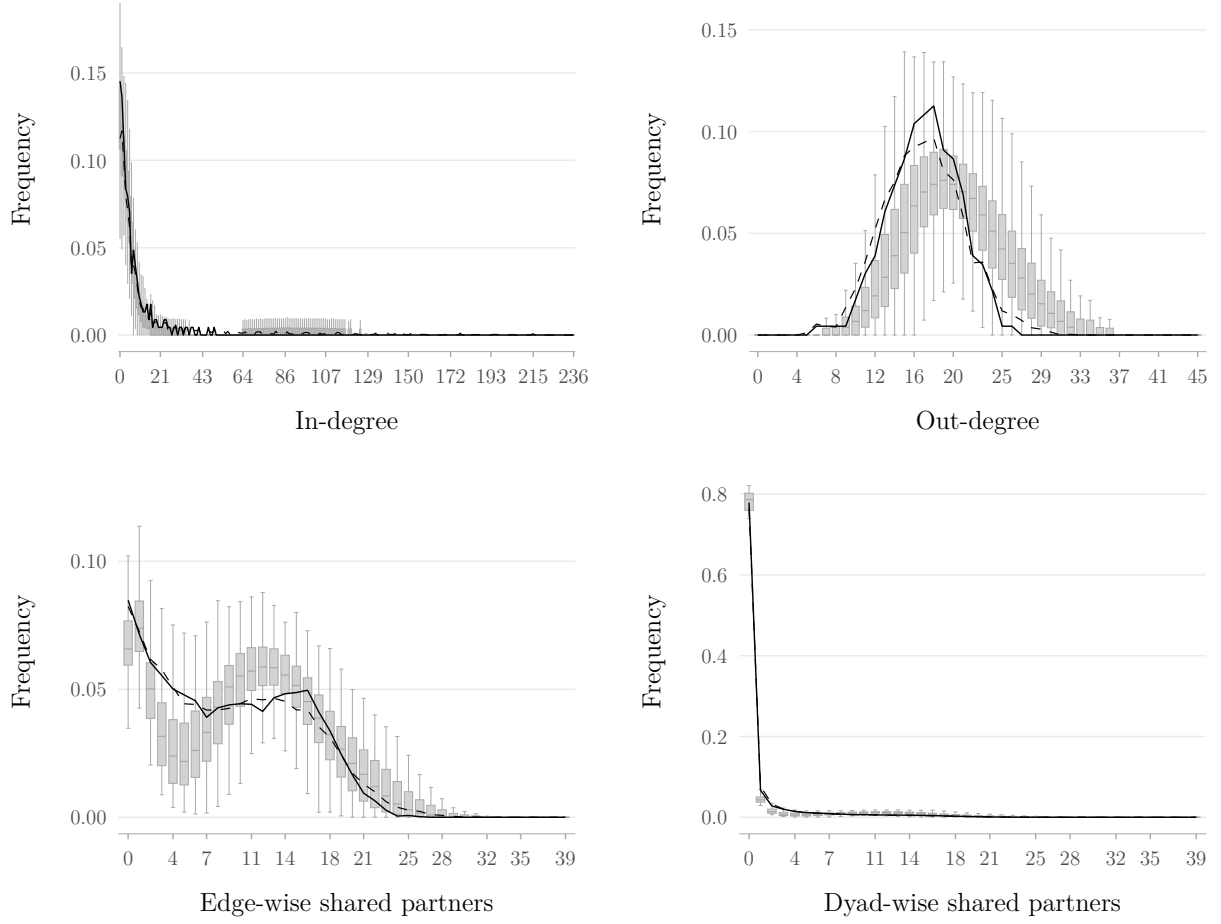


Figure 2: **Goodness of fit for the full model.** Observed statistics (solid: median; dashed: mean) against boxplots of 100 simulated networks per transition, pooled across the eleven transitions. Clockwise from top left: in-degree, out-degree, dyadwise shared partners, edgewise shared partners.

C Pseudolikelihood Estimation: Bias and Interval Coverage

Our TERGM models are estimated by maximum pseudolikelihood with bootstrap confidence intervals (Section A.1). Pseudolikelihood approximates the likelihood from each dyad’s conditional probability given the remainder of the network. This approach can bias the estimates of the endogenous terms and, without the bootstrap, understates their uncertainty (van Duijn et al., 2009; Desmarais and Cranmer, 2012). To check the estimator on these data, we draw 100 sets of eleven networks from the fitted full model, where each transition is simulated conditional on the observed lagged network and covariates of that transition. This procedure treats the fitted model as the data-generating process, in the manner of a parametric bootstrap (Schmid and Desmarais, 2017). We then re-estimate the full specification on each set with bootstrapped pseudolikelihood using $R = 100$ bootstrap replications,

and compare the estimates and 95% confidence intervals with the generating coefficients.

Simulating from the fitted model using `btergm` by default starts the MCMC chains at the observed network and runs them for 10,000 proposals, which on networks of 46,000–89,000 ordered dyads leaves a simulated network sharing most of its ties with the observed one. Run longer, the chains move away from the data and settle after roughly 20–30 million proposals. At that equilibrium, the fitted model has approximately twice as many ties as the observed networks (1.6 to 3.2 times, by transition), 4.6 times as many mutual dyads, and 2.3 times the edgewise shared-partner statistic, while the geometrically weighted in-degree and non-edgewise shared-partner statistics remain within 10 percent of the observed values (Table 8). This qualifies Figure 2, which compares the observed networks with short chains from them rather than with draws from the model’s stationary distribution. Pseudolikelihood coefficients need not satisfy the moment conditions imposed by maximum likelihood, so under strong dependence the model can predict each tie well given the rest of the network yet still generate networks that differ from the data. For the bias check we draw from the equilibrium, using a burn-in of 30 million proposals. We then continue the chain for 100 million proposals, keeping one network every million, yielding 100 draws per transition. The resulting simulated networks are denser than the observed data. The check below assesses whether the estimator recovers the generating coefficients on networks of this size and covariate structure at these parameter values. However, it does not establish that the estimator would recover the maximum-likelihood coefficients of a model centered on the observed networks; this cannot be simulated without those estimates.

	Observed	Simulated	Ratio
Edges	4,304	8,661	2.04
Mutual dyads	505	2,141	4.60
Persisting ties	1,442	2,072	1.44
GW edgewise shared partners	5,676	12,888	2.31
GW in-degree	291	310	1.07
GW non-edgewise shared partners	13,231	12,027	0.93

Table 8: **Observed statistics against draws from the fitted model’s equilibrium.** Means over the eleven transitions of the observed statistic and of the mean over 100 simulated networks per transition (burn-in of 30 million proposals, draws one million proposals apart); the ratio is the mean over transitions of simulated to observed.

	Generating value	Mean estimate	Bias	Bias/SD	Width ratio	Coverage (%)
Edges	-7.349	-7.343	0.006	0.03	1.08	94
Reciprocity	3.261	3.264	0.003	0.11	1.07	93
Tie Persistence ($t - 1$)	0.485	0.487	0.002	0.15	1.03	93
Triadic Closure	1.112	1.095	-0.017	-0.16	0.96	87
GW In-degree	-0.430	-0.477	-0.047	-0.22	0.98	95
Open Triads	-0.232	-0.234	-0.002	-0.26	1.16	96
GW Out-degree	111.170	684.212	573.042	0.83	2.15	31
Party Hom. $\times$ Time	0.019	0.019	0.001	0.18	1.04	92
Linear Time Trend	-0.032	-0.032	0.000	-0.05	1.05	94
Speech Vol. (Pop.)	0.578	0.579	0.000	0.06	1.05	90
Speech Vol. (Soc.)	0.032	0.033	0.001	0.12	1.03	91
Maj. Leader Hom.	-0.038	-0.036	0.003	0.08	0.89	88
Maj. Leader (Pop.)	0.103	0.104	0.001	0.02	0.97	90
Min. Leader Hom.	0.036	0.038	0.001	0.05	1.06	91
Min. Leader (Pop.)	0.101	0.103	0.001	0.04	1.03	95
Comm. Chair Hom.	0.034	0.036	0.002	0.08	1.07	93
Comm. Chair (Pop.)	0.136	0.139	0.003	0.07	0.90	91
Party Hom.	-0.007	-0.008	-0.002	-0.06	1.12	91
DW-NOM 1 (Dist.)	-0.320	-0.320	0.000	0.00	0.97	91
CF Score (Dist.)	-0.213	-0.211	0.001	0.08	1.09	93
Seniority (Dist.)	-0.010	-0.010	0.000	-0.03	0.92	87
Gender Hom.	-0.014	-0.012	0.002	0.15	1.03	90
Race Hom. (Black)	-0.013	-0.013	0.000	-0.01	1.04	94
Eth. Hom. (Latino)	0.046	0.046	0.000	-0.01	0.95	90
Urban. (Dist.)	-0.192	-0.191	0.002	0.05	0.90	93
Comm. Hom.	0.069	0.070	0.001	0.09	0.97	89

Table 9: **Bias and interval coverage of the bootstrapped pseudolikelihood estimator.** 100 sets of eleven networks drawn from the fitted full model, each refit with the full specification ($R = 100$). Bias is the mean estimate subtracted by the generating value; Bias/SD scales it by the standard deviation of the estimates across replicates; the width ratio divides the mean 95% bootstrap interval width by the width of the empirical 95% range of the estimates (1 indicates intervals of the right size); coverage is the share of replicates whose interval contains the generating value (nominal 95). The GW out-degree term’s change statistic is saturated in these networks, so its coefficient is not identified (Section A.1).

Table 9 reports the result of this procedure. For the 25 identified terms, the point estimates are essentially unbiased. The largest absolute bias is 0.047 log-odds, which corresponds to the geometrically weighted in-degree term (-0.477 vs a generating value of -0.430). No term’s bias exceeds 0.26 standard deviations of its sampling distribution, and 17 terms fall below 0.1. The mean widths of the bootstrap intervals fall between 0.89 and 1.16 times the empirical 95 percent range of the estimates, and their coverage ranges from 87 to 96 percent (mean 92) against a nominal 95 percent. The shortfall of a few percentage points here is within the Monte Carlo error of 100 replicates. The one exception is the geometri-

cally weighted out-degree term, which has a mean estimate of 684 relative to a generating value of 111, and only 31 percent coverage. In these networks, this term’s change statistic is saturated; every member’s out-degree exceeds the point at which the term stops changing. Therefore, this coefficient is not identified by the data and its scale is arbitrary (Section A.1). We report it for completeness and decline to interpret it. Under the fitted model, the bootstrapped pseudolikelihood estimator recovers its own coefficients, which supports reading the estimates in Table 2 as estimates of a well-defined quantity, even though the model does not reproduce the density of the observed networks at equilibrium.

D Tie Prediction: In-Sample and Out-of-Sample

Figure 3 reports receiver-operating-characteristic (ROC) and precision-recall (PR) curves for tie prediction. Curves are computed separately for each transition, then the areas are averaged. In sample, the full model attains a mean AUC-ROC of 0.906 and a mean area under the precision-recall curve (AUC-PR) of 0.559. For the out-of-sample test we re-estimate the model on the transitions into the 105th–113th Congresses only and predict ties in the held-out 114th and 115th: AUC-ROC is **0.921** and AUC-PR is **0.604**. Because the networks are sparse, precision-recall is a conservative metric in that the PR baseline (the precision of a random predictor) is close to the density of the network being predicted. Here, the random-predictor baseline averages 0.070 in sample and 0.064 out of sample, so the model’s out-of-sample AUC-PR represents roughly a nine-fold improvement over chance. The out-of-sample specification omits the time-trend terms, whose index would restart on the held-out window. Predictive performance on unseen Congresses indicates that the estimated structure generalizes rather than merely absorbing within-sample noise.

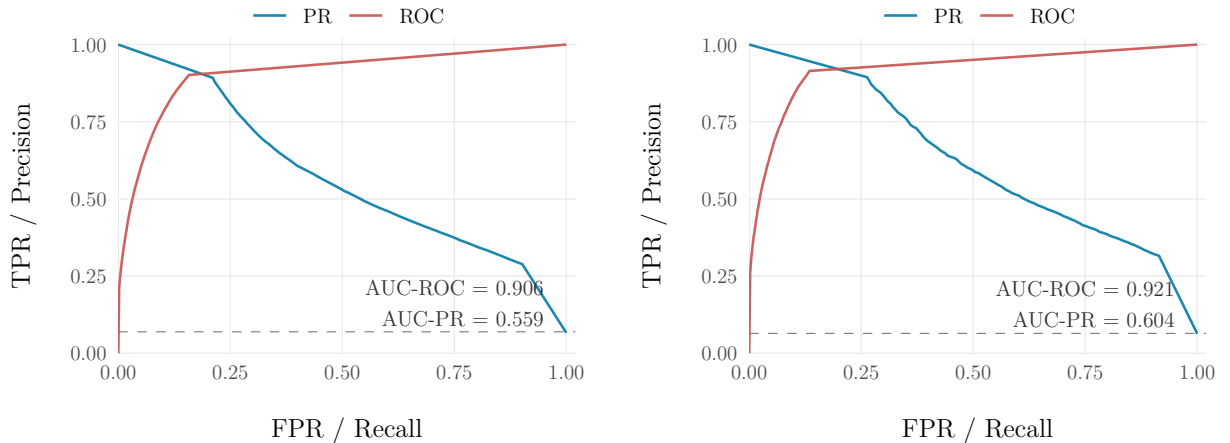


Figure 3: **Tie prediction.** ROC (red) and precision-recall (blue) curves, one per transition; reported areas are averages across transitions. The dashed grey line is the random-predictor precision baseline (0.070 in sample, 0.064 out of sample), close to the average density of the networks being predicted. Left: in-sample, full model, all eleven transitions. Right: out-of-sample, model estimated through the 113th Congress and evaluated on the 114th–115th.

Table 10 reports the coefficients of the training fit beside the full model. The substantive

terms retain their sign when the estimation window is truncated, and the structural terms are essentially unchanged. At $R = 100$ on nine rather than eleven transitions, the urbanization term decreases in magnitude (-0.19 to -0.09) and it loses significance along with the small sender-volume term. The largest movement among the interpreted structural terms is the geometrically weighted in-degree term, which strengthens from -0.43 to -0.53 , with each estimate falling inside the other’s interval. Taken together, these results suggest the full-sample estimates are not driven by the held-out Congresses.

	Full (all transitions)	Training fit (105th–113th)
Edges	-7.349 ($-7.946; -6.715$)*	-7.077 ($-7.768; -6.687$)*
Reciprocity	3.261 ($3.034; 3.477$)*	3.317 ($3.034; 3.553$)*
Triadic Closure	1.112 ($1.026; 1.184$)*	1.080 ($1.007; 1.132$)*
Open Triads	-0.232 ($-0.250; -0.217$)*	-0.245 ($-0.258; -0.231$)*
Tie Persistence ($t - 1$)	0.485 ($0.418; 0.546$)*	0.482 ($0.403; 0.562$)*
Linear Time Trend	-0.032 ($-0.087; -0.001$)*	
Party Hom.	-0.007 ($-0.283; 0.203$)	-0.008 ($-0.120; 0.127$)
Party Hom. $\times$ Time	0.019 ($0.004; 0.049$)*	
DW-NOM 1 (Dist.)	-0.320 ($-0.443; -0.185$)*	-0.283 ($-0.445; -0.110$)*
CF Score (Dist.)	-0.213 ($-0.273; -0.141$)*	-0.281 ($-0.342; -0.212$)*
Speech Vol. (Pop.)	0.578 ($0.531; 0.629$)*	0.547 ($0.498; 0.608$)*
Speech Vol. (Soc.)	0.032 ($0.008; 0.063$)*	0.029 ($-0.005; 0.059$)
Seniority (Dist.)	-0.010 ($-0.013; -0.006$)*	-0.015 ($-0.018; -0.008$)*
Urban. (Dist.)	-0.192 ($-0.330; -0.056$)*	-0.090 ($-0.272; 0.042$)
Comm. Hom.	0.069 ($0.043; 0.097$)*	0.064 ($0.038; 0.091$)*
Gender Hom.	-0.014 ($-0.052; 0.019$)	-0.016 ($-0.069; 0.019$)
Race Hom. (Black)	-0.013 ($-0.075; 0.062$)	-0.037 ($-0.076; 0.019$)
Eth. Hom. (Latino)	0.046 ($-0.040; 0.148$)	0.064 ($-0.029; 0.154$)
Maj. Leader Hom.	-0.038 ($-0.102; 0.021$)	-0.029 ($-0.102; 0.049$)
Min. Leader Hom.	0.036 ($-0.015; 0.090$)	0.032 ($-0.036; 0.086$)
Comm. Chair Hom.	0.034 ($-0.024; 0.092$)	0.031 ($-0.047; 0.087$)
Maj. Leader (Pop.)	0.103 ($-0.060; 0.284$)	0.139 ($-0.064; 0.403$)
Min. Leader (Pop.)	0.101 ($-0.116; 0.357$)	0.044 ($-0.149; 0.311$)
Comm. Chair (Pop.)	0.136 ($-0.037; 0.271$)	0.181 ($-0.039; 0.347$)

* Null hypothesis value outside the confidence interval.

Table 10: **Out-of-sample training fit.** Full specification without the time-trend terms (their index would restart on the held-out window), estimated on the transitions into the 105th–113th Congresses only ($R = 100$; point estimates are the quantity of interest for prediction). 95% bootstrap confidence intervals in parentheses.

E External Validation

In this section, we test whether discursive alignment at Congress $t - 1$ predicts a co-sponsorship tie at t , conditional on co-sponsorship at $t - 1$ and the full set of covariates. Note that the design fixes only the temporal ordering, so the estimate is an association, not a causal effect.

Co-sponsorship networks. We construct House bill-co-sponsorship networks for the 104th–115th Congresses as weighted bipartite projections of the member–bill co-sponsorship incidence matrix, using Fowler’s co-sponsorship data (Fowler, 2006) through the 107th Congress and govinfo bulk records from the 108th onward. On the 108th, which both sources cover, the two weighted projections are perfectly collinear at $r = 1.000$ across all dyads among the 436 members common to both. We binarize each projection by extracting its backbone under the stochastic-degree-sequence model at $\alpha = 0.0001$ (Neal, 2014). We apply the extraction to the full chamber before restricting to the speech networks’ node sets so that the null model conditions on each member’s chamber-wide co-sponsorship activity. Node matching is complete in eleven of twelve Congresses and covers 383 of 385 members in the 113th. Resulting densities rise from roughly 0.072 in the 104th to 0.178 in the 115th, peaking at 0.190 in the 114th.

Specification. The model mirrors the specification of Table 2, adapted to an undirected outcome over the eleven transitions from the 105th to the 115th Congress. The mutuality term is undefined for undirected networks and dropped. The separate in- and out-degree terms collapse to a single geometrically weighted degree term with decay $\phi = 0.3$. Receiver/sender covariates become node covariates, and co-sponsorship at $t - 1$ is included in the model as an edge covariate. Here, the predictor of interest is the lagged speech tie, symmetrized so that a dyad is considered aligned regardless of the direction alignment ran. For this specification, we use the same decay parameters as the main model of Table 2.

Results. At $R = 1000$, the lagged alignment coefficient is 0.181 (95% CI [0.108, 0.246]), a multiplicative increase of roughly 20% in the odds of a co-sponsorship tie, controlling for prior co-sponsorship, ideological distance, shared committee membership, and all other covariates. The model predicts ties with an AUC-ROC of 0.938 and an AUC-PR of 0.771, roughly four times the random-predictor baseline of 0.191. A coordinate-wise sweep over ten configurations of the three decay parameters results in estimates ranging from 0.169 to 0.184. In each of these configurations, the estimates are statistically significant. Pooled miss rates range from 0.232 to 0.278, with three configurations tied at the minimum—including the final co-sponsorship model reported in this section. Logged speaking volume is not statistically significant in this model (-0.007 , $[-0.032, 0.017]$), whereas in the speech networks the receiver effect for logged speaking volume is the largest exogenous coefficient (0.578). Speaking more raises the odds that a member’s language stands out to colleagues, but not the odds that a pair co-sponsors. This suggests the alignment estimate does not simply reflect a common dependence of the two networks on speaking activity.

Table 11: **Co-sponsorship prediction.** Bootstrapped TERGM over the eleven transitions from the 105th to the 115th Congress, $R = 1000$. 95% bootstrap confidence intervals in brackets. The discursive-alignment term is the lagged, symmetrized speech tie.

	Co-sponsorship
Edges	-2.898 [-3.832; -2.144]*
Co-sponsorship ($t - 1$)	1.987 [1.909; 2.108]*
Discursive Alignment ($t - 1$)	0.181 [0.108; 0.246]*
Triadic Closure	1.298 [1.119; 1.507]*
Degree Concentration	-0.502 [-0.983; 0.253]
Open Triads	-0.227 [-0.256; -0.200]*
Party Hom. $\times$ Time	0.006 [-0.083; 0.088]
Linear Time Trend	0.006 [-0.074; 0.108]
Speech Vol. (Node)	-0.007 [-0.032; 0.017]
Maj. Leader Hom.	0.629 [0.253; 1.054]*
Maj. Leader (Node)	0.203 [-0.123; 0.580]
Min. Leader Hom.	-0.031 [-0.143; 0.118]
Min. Leader (Node)	0.105 [-0.095; 0.335]
Comm. Chair Hom.	0.242 [0.092; 0.418]*
Comm. Chair (Node)	-0.359 [-0.576; -0.126]*
Party Hom.	-0.717 [-0.957; -0.201]*
DW-NOM 1 (Dist.)	-2.670 [-3.135; -2.059]*
CF Score (Dist.)	-0.457 [-0.612; -0.308]*
Seniority (Dist.)	-0.000 [-0.012; 0.008]
Gender Hom.	-0.105 [-0.173; -0.006]*
Race Hom. (Black)	0.439 [0.322; 0.617]*
Eth. Hom. (Latino)	0.094 [-0.049; 0.319]
Urban. (Dist.)	-1.586 [-1.900; -1.228]*
Comm. Hom.	0.521 [0.454; 0.593]*
Num. obs.	709388

* 0 outside the confidence interval.

Distinctness of the speech and co-sponsorship networks. Edge overlap between the speech and co-sponsorship networks is low, with Jaccard coefficients ranging from 0.08 to 0.15 across Congresses (mean 0.12). The correlation between a member’s degree in the two networks ranges from -0.04 to 0.25 , with a mean of 0.11 . To put these values in context, two reimplementations of the lexical alignment measure overlap at 0.82 , and the lexical and embedding constructions detailed in Section F overlap at 0.33 . Furthermore, the two networks respond to different member attributes. Within the co-sponsorship model, distance in ideological extremity is the largest attribute-based effect (-2.670), followed by urbanization distance (-1.59). Both of these terms are comparatively minor in the speech network model (-0.32 and -0.19 , respectively). Because the two outcomes have different baseline densities, we contrast the ordering of the effects rather than their magnitudes.

Together with the alignment estimate reported above, these results indicate that discursive alignment encodes predictive information that co-sponsorship structure alone does not.

F Measurement Robustness

The networks analyzed in the main text are constructed using TF-IDF and cosine similarity, thresholded row-wise. To assess whether the substantive conclusions depend on this method of network construction, we rebuilt the networks from the same one-minute corpus using two approaches and re-estimated the full specification on each.

Reimplementation of the lexical TF-IDF pipeline. To establish a baseline for comparison, we reimplemented the original lexical network construction pipeline. We replicated the construction from scratch, including concatenation, preprocessing, TF-IDF, cosine similarity, and row-wise thresholding based on z -score. This reimplementation reproduces important global, nodal, and dyadic properties of the original networks. Global sparsity is preserved, with densities matching the original values within 0.002. At the node level, in-degree distributions correlate at $r = 0.99$, maintaining the relative prominence of individuals. Finally, at the dyadic level, the edge Jaccard similarity between the networks falls between 0.80 and 0.83 (Table 12). The residual edge disagreement, rather than representing measurement disagreement, reflects near-threshold values crossing the z cutoff under small differences in vocabulary. When applying hard thresholding to a continuous distribution, minor computational variations in text preprocessing can lead to near-threshold dyads crossing the rigid z -score boundary. Stored and reimplemented edge-level cosine similarities correlate at $r = 0.98$ – 0.99 , with a mean absolute difference of 0.014.

Sentence-embedding similarity. As a distinct measure, we embed each speech with a sentence-transformer model (all-mpnet-base-v2). Embeddings are averaged within member-Congress pairs to create a mean vector for each pair, before calculating cosine similarities between members’ mean vectors for each Congress. The resulting similarity rows depart sharply from normality and exhibit skewness in the opposite direction from the lexical rows (Figure 4). TF-IDF rows are right-skewed (pooled median row skewness +0.67), while embedding rows are markedly left-skewed (−0.65), with most colleagues clustered at high semantic similarity and a long lower tail. That lower tail inflates each row’s standard deviation, so the $z > 1.64$ rule—which presumes approximate normality within rows—selects far fewer than its nominal share of ties and would produce networks 2.5 to 10.5 times sparser (5.2 on average) than the lexical ones (Figure 4, bottom panel). To separate the effect of the similarity measure from the confounding effect of network sparsity, we replace the standardized z -score cutoff with a density-matched top- k threshold. For each Congress, we take each row’s top k ties, with k set to match the lexical network’s density in each Congress, holding the tie budget fixed so that differences in the estimates reflect the similarity measure rather than sparsity. The resulting networks differ genuinely from the lexical ones (edge Jaccard ≈ 0.33 , against 0.82 between the two lexical implementations) while preserving the in-degree ordering ($r \approx 0.81$). Table 12 reports both comparisons by Congress.

Embedding model and aggregation. Each one-minute speech is encoded with the all-mpnet-base-v2 sentence transformer,⁴ which builds on Microsoft’s MPNet pre-training architecture (Song et al., 2020) within the Sentence-BERT framework (Reimers and Gurevych, 2019) and is fine-tuned on over one billion sentence pairs for the task of producing semantically meaningful sentence-level embeddings. The model maps variable-length text into a fixed 768-dimensional dense vector space in which geometric proximity reflects semantic similarity. Two speeches that frame an issue similarly lie close together in that space even when they share little vocabulary. Unlike count-based representations like TF-IDF, which aggregate word-level statistics without regard to semantic meaning, embedding networks can recover this semantic dimension, a property that makes them a potentially informative measurement alternative. Speech embeddings are L_2 -normalized and averaged within each member-Congress pair. Because the arithmetic mean of unit vectors is not itself unit-length, these averaged vectors are then re-normalized so that the cosine similarity between two members’ vectors reduces to a dot product. Similarities are then computed for all dyads within each Congress on the same member sets as the lexical networks. Unlike the lexical pipeline, the similarity rows are not standardized to z -scores and thresholded. Instead, we employ the density-matched top- k rule described above.

⁴Model card: <https://huggingface.co/sentence-transformers/all-mpnet-base-v2>.

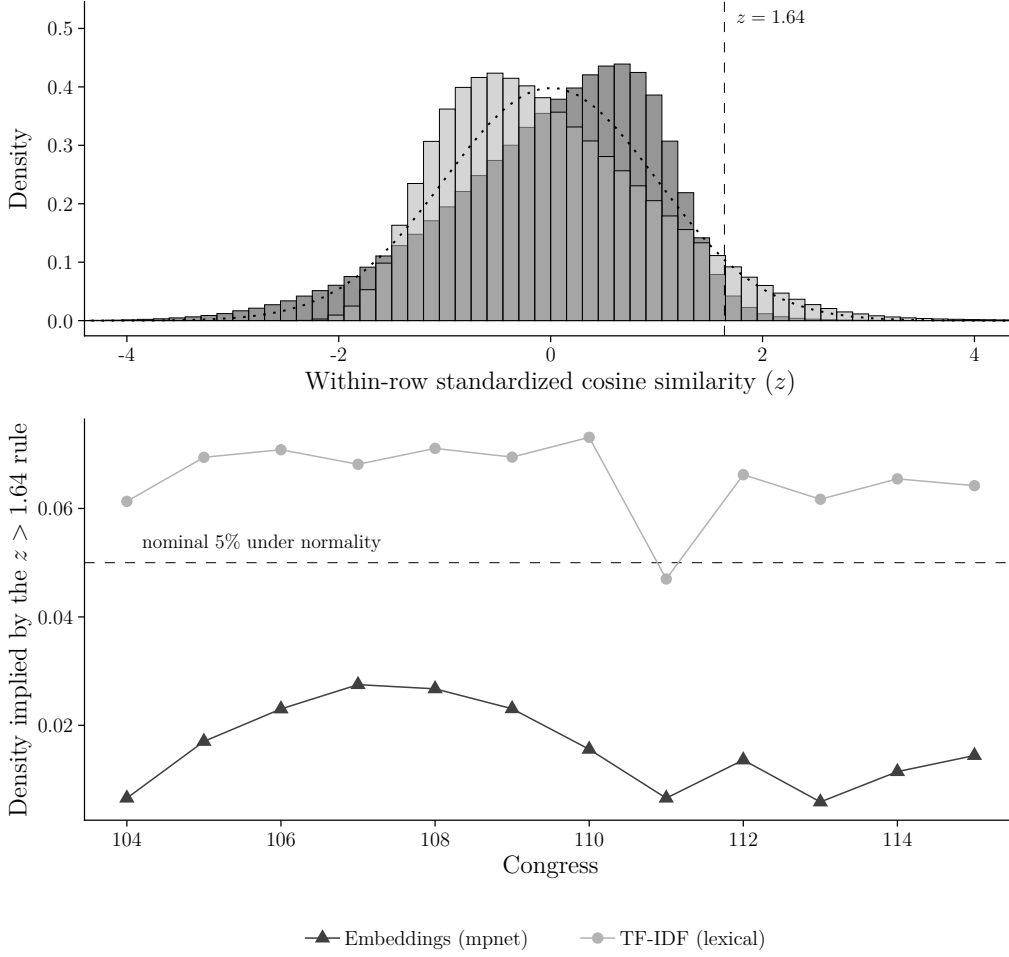


Figure 4: **Skewness of cosine similarity and the limitation of standardized thresholding.** Top: distribution of within-row standardized cosine similarities, pooled across all member-rows of the 104th–115th Congresses, against the standard normal distribution (dotted). TF–IDF rows are right-skewed (median row skewness +0.67), with 6.5% of values exceeding $z = 1.64$, near the nominal 5 percent. Embedding rows are left-skewed (−0.65), with most colleagues sitting above the row mean. The long lower tail inflates the row standard deviation, so only 1.5 percent of values exceed the cutoff. Bottom: the network density each Congress would receive under the row-wise $z > 1.64$ rule; the embedding networks are 2.5–10.5 times sparser (mean 5.2), motivating the density-matched top- k thresholding used in the embedding-based specification. The TF–IDF series uses the reimplemented similarity matrices (the original pipeline’s dense matrices are not preserved; its edge-level cosines are highly correlated with the reimplementations at $r = 0.98$ – 0.99).

Congress	n	Density	Reimpl. vs. original		Embeddings vs. original	
			In-deg. r	Jaccard	In-deg. r	Jaccard
104	386	0.060	0.994	0.83	0.80	0.33
105	352	0.068	0.995	0.81	0.81	0.32
106	335	0.070	0.996	0.83	0.79	0.34
107	322	0.069	0.994	0.81	0.82	0.32
108	312	0.071	0.996	0.83	0.83	0.34
109	314	0.070	0.995	0.82	0.83	0.37
110	335	0.073	0.995	0.83	0.85	0.36
111	387	0.045	0.995	0.80	0.81	0.29
112	349	0.067	0.992	0.80	0.83	0.33
113	385	0.061	0.992	0.82	0.75	0.29
114	371	0.065	0.994	0.82	0.77	0.31
115	374	0.064	0.994	0.82	0.81	0.32
Mean		0.065	0.994	0.82	0.81	0.33

Table 12: **Comparison of alternative network constructions.** Density is that of the original network; each block reports the Pearson correlation of members’ in-degrees and the directed-edge Jaccard overlap against the original networks. The original lexical networks are compared against two alternative constructions, namely an independent reimplementaion of the TF-IDF pipeline and semantic networks derived from transformer-based sentence embeddings. Structural agreement is measured at the node level with Pearson correlation of in-degree distributions, denoted r , and at the dyadic level via edge Jaccard similarity. Member sets are identical across all constructions within each Congress.

Table 13 presents the full specification estimated on the original networks alongside both alternative constructions ($R = 1000$ each). Against the original estimates, the reimplementaion column agrees in sign on all terms and in significance on 25 of 26 (the exception is the chair receiver effect, which is null in the main model). The embedding column agrees in sign on 23 of 26 and in significance on 24 of 26. The disagreements are confined to leadership and demographic terms that are not statistically significant in the main model. All terms that are significant in the main model are same-signed and significant under both alternative constructions.

Turning to the disagreeing terms, two patterns emerge. First, the receiver effect of speaking volume falls by more than half under embeddings (0.58 to 0.25). This behavior is consistent with the volume channel operating largely through lexical breadth, where more speech covers more distinct vocabulary. This inflates lexical similarity but does not necessarily translate to increased meaning-level similarity. Second, two secondary terms that are positive but insignificant in the main model cross into significance under alternative constructions: the chair receiver effect (under both constructions here, and under the union-node-set model of Section A.5) and minority-leader homophily (under embeddings, and under the equal-length reconstruction of Section I.3). These terms are consistently signed but low-powered effects that fall near the significance boundary across specifications, demonstrating substantial measurement sensitivity. Accordingly, we avoid interpreting them substantively.

	Original	TF-IDF	MPNet
Edges	-7.349* [-7.946;-6.715]	-6.923* [-7.464;-6.306]	-5.216* [-5.615;-4.886]
Reciprocity	3.261* [3.034;3.477]	3.407* [3.186;3.652]	2.565* [2.420;2.684]
Tie Persistence ($t - 1$)	0.485* [0.418;0.546]	0.463* [0.387;0.537]	0.362* [0.318;0.403]
Triadic Closure	1.112* [1.026;1.184]	1.077* [0.984;1.159]	1.275* [1.213;1.327]
Popularity	-0.430* [-0.564;-0.239]	-0.380* [-0.490;-0.215]	-0.669* [-0.884;-0.416]
Open Triads	-0.232* [-0.250;-0.217]	-0.229* [-0.249;-0.213]	-0.319* [-0.329;-0.311]
Sociality	111.170* [71.451;180.243]	13.746* [6.017;88.048]	609.302* [473.292;5171.534]
Party Hom. $\times$ Time	0.019* [0.004;0.049]	0.017* [0.005;0.044]	0.009* [0.000;0.022]
Linear Time Trend	-0.032* [-0.087;-0.001]	-0.031* [-0.080;-0.004]	-0.027* [-0.050;-0.013]
Speech Vol. (Pop.)	0.578* [0.531;0.629]	0.539* [0.493;0.584]	0.245* [0.201;0.286]
Speech Vol. (Soc.)	0.032* [0.008;0.063]	0.032* [0.006;0.063]	0.137* [0.116;0.164]
Maj. Leader Hom.	-0.038 [-0.102;0.021]	-0.055 [-0.126;0.006]	-0.060 [-0.152;0.013]
Maj. Leader (Pop.)	0.103 [-0.060;0.284]	0.089 [-0.115;0.264]	-0.029 [-0.146;0.069]
Min. Leader Hom.	0.036 [-0.015;0.090]	0.027 [-0.032;0.096]	0.124* [0.076;0.172]
Min. Leader (Pop.)	0.101 [-0.116;0.357]	0.091 [-0.132;0.328]	0.071 [-0.062;0.219]
Comm. Chair Hom.	0.034 [-0.024;0.092]	0.061 [-0.005;0.125]	0.022 [-0.022;0.065]
Comm. Chair (Pop.)	0.136 [-0.037;0.271]	0.161* [0.015;0.251]	0.183* [0.074;0.278]
Party Hom.	-0.007 [-0.283;0.203]	-0.022 [-0.270;0.160]	-0.071 [-0.185;0.029]
[DW-NOM 1] (Dist.)	-0.320* [-0.443;-0.185]	-0.327* [-0.461;-0.183]	-0.218* [-0.363;-0.067]
CF Score (Dist.)	-0.213* [-0.273;-0.141]	-0.217* [-0.288;-0.143]	-0.192* [-0.248;-0.128]
Seniority (Dist.)	-0.010* [-0.013;-0.006]	-0.011* [-0.015;-0.006]	-0.005* [-0.009;-0.002]
Gender Hom.	-0.014 [-0.052;0.019]	-0.000 [-0.046;0.044]	0.038 [-0.006;0.077]
Race Hom. (Black)	-0.013 [-0.075;0.062]	-0.047 [-0.098;0.017]	0.071 [-0.001;0.148]
Eth. Hom. (Latino)	0.046 [-0.040;0.148]	0.036 [-0.057;0.135]	0.080 [-0.004;0.149]
Urbanization (Dist.)	-0.192* [-0.330;-0.056]	-0.190* [-0.321;-0.058]	-0.376* [-0.495;-0.261]
Comm. Hom.	0.069* [0.043;0.097]	0.077* [0.052;0.105]	0.045* [0.015;0.076]

Table 13: **Measurement robustness of the full specification.** Estimates with 95% bootstrap confidence intervals; * indicates the interval excludes zero. Column 1: original networks. Column 2: independent reimplementations of the TF-IDF/cosine/row- z pipeline. Column 3: sentence-embedding similarity with density-matched top- k thresholding. The geometrically weighted out-degree coefficient is not comparable across thresholding rules (top- k fixes out-degree by construction) and should not be interpreted.

G Threshold Sensitivity

The networks in the main text are created by assigning a tie from member i to j when $z_{ij} > 1.64$. To assess the sensitivity of the network properties and model estimates to this threshold, we use the same cosine similarities to create the networks under three alternative cutoff values: $z_{ij} > 1.28, 1.96$, and 2.33 . These values correspond to the upper 10, 2.5, and 1 percent of the standard normal distribution, against 5 percent at 1.64. The alternative cutoffs are applied to the reimplemented TF-IDF similarity matrices of Section F.

Table 14 displays descriptive summary statistics of the networks at each cutoff. By construction, density directly depends on the cutoff, falling from 0.112 on average at the 1.28 cutoff to 0.021 at 2.33. Transitivity follows a similar pattern as density, falling from 0.38 to 0.20 across the same cutoffs, whereas reciprocity is relatively constant. Moreover, the top decile of members by in-degree receives 51 to 59 percent of all incoming ties on average, and party assortativity is positive in all Congress-cutoff combinations and rises as the cutoff becomes more strict (from 0.24 to 0.42 on average). The minimum assortativity value occurs in the 107th Congress across all four cutoffs.

Network	Density	Reciprocity	Transitivity	Assortativity	Top-decile share
Original, $z > 1.64$	0.065 (0.045–0.073)	0.23 (0.14–0.33)	0.29 (0.24–0.33)	0.30 (0.11–0.45)	0.60 (0.51–0.65)
$z > 1.28$	0.112 (0.099–0.122)	0.30 (0.22–0.41)	0.38 (0.36–0.42)	0.24 (0.09–0.41)	0.51 (0.45–0.56)
$z > 1.64$	0.066 (0.047–0.073)	0.26 (0.15–0.37)	0.30 (0.24–0.33)	0.31 (0.11–0.47)	0.57 (0.49–0.63)
$z > 1.96$	0.039 (0.024–0.047)	0.24 (0.11–0.33)	0.24 (0.17–0.27)	0.38 (0.14–0.52)	0.59 (0.51–0.65)
$z > 2.33$	0.021 (0.010–0.028)	0.24 (0.10–0.33)	0.20 (0.14–0.25)	0.42 (0.17–0.56)	0.56 (0.50–0.61)

Table 14: **Network properties by tie threshold.** Mean across the twelve Congresses, with the range in parentheses. The first row describes the original networks; the others, the reimplemented TF-IDF networks cut at each threshold. Reciprocity is the share of ties that are reciprocated; transitivity is the global clustering coefficient of the undirected network in which i and j are connected if either names the other; assortativity is the directed form of Newman’s nominal coefficient by party; the top-decile share is the share of incoming ties received by the tenth of members with the highest in-degree.

Table 16 compares the results of fitting the main text’s full model specification on each of the alternative-cutoff networks with $R = 100$ bootstrap replications against the main model and the reimplemented 1.64-cutoff networks ($R = 1000$; Table 13). Most substantive findings do not depend on the cutoff. Tie persistence, triadic closure, open triads, the linear time trend, the receiver-volume term, both ideology distances, seniority, urbanization, and shared committee membership retain both their sign and statistical significance at all four cutoff values. However, three estimates do depend on the cutoff. First, the geometrically weighted in-degree term is negative and significant at 1.64 and stricter cutoffs, growing in magnitude as the cutoff tightens ($-0.38, -0.68$, and -0.81), but is positive and significant at 1.28 (0.35). The concentration of incoming ties that remains after conditioning on speaking volume is therefore a property of the networks at the main cutoff and stricter ones. At the least strict cutoff, where the number of ties grows by approximately 70 percent, the coefficient reverses, although the raw concentration in Table 14 remains. Second, the party-by-time interaction is positive at all four cutoffs but only significant in three of them (the exception is 1.96, which

has a coefficient of 0.016). Third, the sender-volume term changes sign across cutoffs. This phenomenon lines up with our expectations from the results of Section A.2, which indicate this term is the least stable across the various decay configurations tested. As in the other robustness models, several leadership, chair, and demographic terms that are null in the main model are statistically significant at individual cutoffs; we decline to interpret them substantively. The geometrically weighted out-degree coefficient is not comparable across cutoffs, since its change statistic depends on the out-degree that each cutoff fixes.

H Reciprocity and Transitivity Against Null Models

Cosine similarity is symmetric, so a pair of members with unusually similar language tends to cross both members’ row thresholds. This produces reciprocated ties despite the fact that each direction is evaluated against a different baseline. To separate this effect of the construction from alignment specific to pairs, we compare the observed reciprocity and transitivity of each network with two null models presented in Table 15. The degree-preserving null model modifies each network by swapping edges in a manner that maintains members’ in- and out-degree. The construction null model rebuilds each similarity matrix from its grand mean μ and an additive effect for each member, $s_{ij} = \mu + a_i + a_j + e_{ij}$, with the residuals e_{ij} shuffled across pairs. We threshold the resulting matrix with the same row-wise rule. This procedure preserves the symmetry of the cosine similarity matrix and each member’s overall similarity to the chamber, and removes the alignment that is specific to pairs. Each null model is averaged over 20 draws per Congress. As in Section G, the comparisons use the reimplemented similarity matrices.

Relative to the degree-preserving null, reciprocity is high. On average, 26 percent of ties are reciprocated, relative to 8 percent when only the degrees are kept. Relative to the construction null, however, reciprocity is low. The construction alone reciprocates 34 percent of ties on average, which is more than observed in all Congress-cutoff combinations. This indicates that the high reciprocity of these networks is a property of the measure (symmetric similarity judged against member-level baselines) rather than evidence of mutual alignment between particular pairs. We therefore interpret the reciprocity coefficient with caution.

Transitivity, on the other hand, behaves differently than reciprocity. The observed networks display higher transitivity than both nulls in all Congresses (0.30 on average, against 0.22 under the degree-preserving null and 0.15 under the construction null). The observed networks are also more transitive than the construction null at all cutoff thresholds. Triadic closure therefore appears to reflect alignment shared within groups of members beyond what the degree distribution or the construction produces.

Congress	Reciprocity			Transitivity		
	Observed	Degree	Construction	Observed	Degree	Construction
104	0.20	0.07	0.28	0.29	0.22	0.15
105	0.24	0.09	0.32	0.30	0.24	0.17
106	0.29	0.08	0.39	0.31	0.23	0.16
107	0.37	0.07	0.44	0.30	0.22	0.16
108	0.31	0.08	0.39	0.32	0.24	0.16
109	0.30	0.08	0.38	0.31	0.23	0.16
110	0.26	0.09	0.33	0.33	0.25	0.16
111	0.15	0.04	0.27	0.24	0.16	0.13
112	0.21	0.08	0.29	0.29	0.23	0.16
113	0.23	0.07	0.30	0.28	0.21	0.15
114	0.27	0.07	0.34	0.29	0.22	0.15
115	0.26	0.07	0.33	0.29	0.21	0.15
Mean	0.26	0.08	0.34	0.30	0.22	0.15
Mean, $z > 1.28$	0.30		0.35	0.38		0.26
Mean, $z > 1.96$	0.24		0.35	0.24		0.09
Mean, $z > 2.33$	0.24		0.38	0.20		0.04

Table 15: **Reciprocity and transitivity against two null models.** Observed values for the reimplemented TF-IDF networks at $z > 1.64$, beside the mean over 20 draws of a degree-preserving null (edge swaps that keep each member’s in- and out-degree) and of a construction null (additive member effects plus shuffled pair residuals, cut with the same row-wise rule). The last three rows give the means at the alternative cutoffs (construction null only). Reciprocity is the share of ties that are reciprocated; transitivity is the global clustering coefficient of the undirected network.

	Original	Reimplemented TF-IDF networks			
	$z > 1.64$	$z > 1.28$	$z > 1.64$	$z > 1.96$	$z > 2.33$
Edges	-7.349* [-7.95, -6.71]	-7.013* [-7.72, -6.27]	-6.923* [-7.46, -6.31]	-6.709* [-7.28, -6.19]	-6.425* [-6.95, -5.82]
Reciprocity	3.261* [3.03, 3.48]	2.970* [2.73, 3.20]	3.407* [3.19, 3.65]	3.545* [3.35, 3.72]	3.715* [3.60, 3.82]
Tie Persistence ($t - 1$)	0.485* [0.42, 0.55]	0.484* [0.42, 0.57]	0.463* [0.39, 0.54]	0.544* [0.47, 0.62]	0.678* [0.58, 0.79]
Triadic Closure	1.112* [1.03, 1.18]	0.990* [0.88, 1.12]	1.077* [0.98, 1.16]	1.130* [1.07, 1.17]	1.231* [1.17, 1.30]
Popularity	-0.430* [-0.56, -0.24]	0.354* [0.12, 0.59]	-0.380* [-0.49, -0.21]	-0.684* [-0.75, -0.55]	-0.810* [-1.00, -0.54]
Open Triads	-0.232* [-0.25, -0.22]	-0.193* [-0.21, -0.18]	-0.229* [-0.25, -0.21]	-0.252* [-0.28, -0.23]	-0.253* [-0.27, -0.24]
Sociality	111.170* [71.45, 180.24]	-7.577 [-31.68, 242.31]	13.746* [6.02, 88.05]	1.083 [-0.50, 8.29]	0.048 [-0.46, 2.07]
Party Hom. $\times$ Time	0.019* [0.00, 0.05]	0.023* [0.00, 0.04]	0.017* [0.00, 0.04]	0.016 [-0.00, 0.05]	0.032* [0.02, 0.06]
Linear Time Trend	-0.032* [-0.09, -0.00]	-0.033* [-0.07, -0.01]	-0.031* [-0.08, -0.00]	-0.042* [-0.11, -0.00]	-0.068* [-0.14, -0.03]
Speech Vol. (Pop.)	0.578* [0.53, 0.63]	0.643* [0.56, 0.71]	0.539* [0.49, 0.58]	0.527* [0.50, 0.56]	0.603* [0.56, 0.66]
Speech Vol. (Soc.)	0.032* [0.01, 0.06]	-0.074* [-0.12, -0.02]	0.032* [0.01, 0.06]	0.012 [-0.02, 0.04]	-0.173* [-0.25, -0.10]
Maj. Leader Hom.	-0.038 [-0.10, 0.02]	0.007 [-0.06, 0.05]	-0.055 [-0.13, 0.01]	-0.089 [-0.18, 0.04]	-0.061 [-0.17, 0.04]
Maj. Leader (Pop.)	0.103 [-0.06, 0.28]	0.337* [0.07, 0.55]	0.089 [-0.11, 0.26]	-0.154 [-0.34, 0.04]	0.027 [-0.17, 0.30]
Min. Leader Hom.	0.036 [-0.01, 0.09]	-0.008 [-0.05, 0.04]	0.027 [-0.03, 0.10]	0.036 [-0.01, 0.12]	-0.032 [-0.15, 0.08]
Min. Leader (Pop.)	0.101 [-0.12, 0.36]	0.041 [-0.18, 0.27]	0.091 [-0.13, 0.33]	0.144* [0.00, 0.33]	0.005 [-0.17, 0.20]
Comm. Chair Hom.	0.034 [-0.02, 0.09]	0.048* [0.00, 0.12]	0.061 [-0.01, 0.13]	0.058 [-0.01, 0.15]	-0.005 [-0.07, 0.09]
Comm. Chair (Pop.)	0.136 [-0.04, 0.27]	0.190* [0.01, 0.41]	0.161* [0.02, 0.25]	0.052 [-0.10, 0.14]	0.079 [-0.23, 0.29]
Party Hom.	-0.007 [-0.28, 0.20]	0.085 [-0.12, 0.35]	-0.022 [-0.27, 0.16]	-0.053 [-0.32, 0.21]	-0.121 [-0.31, 0.06]
DW-NOM 1 (Dist.)	-0.320* [-0.44, -0.19]	-0.317* [-0.44, -0.18]	-0.327* [-0.46, -0.18]	-0.458* [-0.60, -0.29]	-0.500* [-0.62, -0.38]
CF Score (Dist.)	-0.213* [-0.27, -0.14]	-0.115* [-0.19, -0.01]	-0.217* [-0.29, -0.14]	-0.233* [-0.32, -0.12]	-0.248* [-0.36, -0.15]
Seniority (Dist.)	-0.010* [-0.01, -0.01]	-0.009* [-0.01, -0.00]	-0.011* [-0.02, -0.01]	-0.011* [-0.02, -0.00]	-0.009* [-0.01, -0.00]
Gender Hom.	-0.014 [-0.05, 0.02]	0.005 [-0.03, 0.04]	-0.000 [-0.05, 0.04]	-0.019 [-0.08, 0.04]	-0.015 [-0.09, 0.06]
Race Hom. (Black)	-0.013 [-0.07, 0.06]	-0.036* [-0.07, -0.01]	-0.047 [-0.10, 0.02]	-0.031 [-0.08, 0.05]	0.007 [-0.05, 0.09]
Eth. Hom. (Latino)	0.046 [-0.04, 0.15]	0.065 [-0.03, 0.18]	0.036 [-0.06, 0.14]	-0.018 [-0.09, 0.07]	-0.008 [-0.10, 0.09]
Urbanization (Dist.)	-0.192* [-0.33, -0.06]	-0.248* [-0.42, -0.13]	-0.190* [-0.32, -0.06]	-0.306* [-0.49, -0.20]	-0.334* [-0.60, -0.17]
Comm. Hom.	0.069* [0.04, 0.10]	0.052* [0.03, 0.08]	0.077* [0.05, 0.11]	0.078* [0.05, 0.11]	0.068* [0.03, 0.10]

Table 16: **The full specification at alternative tie thresholds.** Column 1: the main model on the original networks ($z > 1.64$, $R = 1000$). Columns 2–5: the reimplemented TF-IDF networks cut at each threshold ($R = 100$, except $R = 1000$ at $z > 1.64$, as in Table 13). 95% bootstrap confidence intervals in brackets below each estimate; * indicates the interval excludes zero.

I Speaking Propensity

This section reports the diagnostics behind the speaking-propensity discussion in the main text.

I.1 Volume and degree

A member’s raw in-degree correlates strongly with how much they speak. The pooled Spearman correlation between in-degree and one-minute speech count is 0.73 (ranging 0.70–0.86 by Congress; 0.74 using total word volume), while the corresponding out-degree correlation is only 0.20. Table 17 reports the correlations by Congress. This asymmetry suggests volume operates through lexical coverage, where speaking more makes a member’s language salient in many colleagues’ similarity distributions, but does not proportionally expand whose language stands out to them. The main specification therefore includes each member’s logged one-minute word volume as receiver and sender covariates. Both terms are positive and significant, mirroring the raw asymmetry (0.58 versus 0.03 in log-odds). The volume terms also quantify how much of the apparent in-degree concentration reflects speaking activity. In an otherwise identical specification without them, the geometrically weighted in-degree coefficient is -1.70 , versus -0.43 in the full model—roughly three-quarters of the apparent concentration is attributable to volume, while the remaining quarter is a residual.

Congress	n	In-deg. $\times$ count	Out-deg. $\times$ count	In-deg. $\times$ words	Out-deg. $\times$ words
104	386	0.82	0.29	0.83	0.26
105	352	0.80	0.40	0.81	0.33
106	335	0.76	0.31	0.78	0.31
107	322	0.72	0.13	0.75	0.11
108	312	0.70	0.25	0.68	0.26
109	314	0.71	0.47	0.74	0.46
110	335	0.72	0.46	0.73	0.46
111	387	0.80	-0.27	0.82	-0.25
112	349	0.82	0.31	0.81	0.29
113	385	0.86	0.09	0.85	0.07
114	371	0.80	0.20	0.79	0.17
115	374	0.80	0.22	0.81	0.21
Pooled	4222	0.73	0.20	0.74	0.19

Table 17: **Speaking volume and network degree.** Spearman correlations between members’ network degree and their one-minute speaking volume (speech count and total word count), by Congress and pooled.

I.2 Functional form of the volume control

The main specification includes each member’s logged word volume linearly as a receiver and a sender term. Because the geometrically weighted in-degree estimate depends heavily on whether volume is controlled, we re-estimated the chosen specification at $R = 100$ bootstrap

replications with three alternative functional forms of the control. In each, the term is included as both receiver and sender effects. The three alternatives we check are logged word volume together with its centered square, a natural cubic spline in logged word volume with three degrees of freedom, and the logged number of one-minute speeches in place of word volume. Table 18 reports the terms that do not involve volume. The geometrically weighted in-degree coefficient is negative and significant under each of the three alternatives. Notably, the term is larger in magnitude under the alternatives (-0.59 , -0.61 , and -0.62 against -0.43 under the linear term). This suggests that the residual concentration of incoming ties does not depend on the linear form.

	Log words (main)	+ Squared	Spline (3 df)	Log speeches	No volume
Popularity (GW in-degree)	-0.430^* $[-0.58, -0.23]$	-0.588^* $[-0.72, -0.36]$	-0.606^* $[-0.75, -0.35]$	-0.615^* $[-0.75, -0.44]$	-1.700^* $[-1.86, -1.51]$
Reciprocity	3.261^* $[3.03, 3.43]$	3.231^* $[3.01, 3.40]$	3.233^* $[3.00, 3.40]$	3.196^* $[2.97, 3.36]$	3.018^* $[2.82, 3.18]$
Triadic closure	1.112^* $[1.03, 1.19]$	1.115^* $[1.03, 1.20]$	1.119^* $[1.03, 1.20]$	1.124^* $[1.04, 1.20]$	1.247^* $[1.15, 1.35]$
Open triads	-0.232^* $[-0.25, -0.21]$	-0.233^* $[-0.25, -0.21]$	-0.235^* $[-0.26, -0.22]$	-0.233^* $[-0.25, -0.21]$	-0.278^* $[-0.29, -0.26]$
Tie persistence ($t - 1$)	0.485^* $[0.42, 0.55]$	0.477^* $[0.41, 0.54]$	0.475^* $[0.41, 0.54]$	0.496^* $[0.44, 0.56]$	0.823^* $[0.78, 0.87]$
Party hom. $\times$ time	0.019^* $[0.00, 0.05]$	0.019^* $[0.00, 0.05]$	0.019^* $[0.00, 0.05]$	0.018^* $[0.00, 0.05]$	0.011 $[-0.00, 0.04]$
DW-NOM 1 (dist.)	-0.320^* $[-0.48, -0.17]$	-0.326^* $[-0.49, -0.18]$	-0.327^* $[-0.49, -0.18]$	-0.340^* $[-0.51, -0.18]$	-0.243^* $[-0.43, -0.10]$
CF score (dist.)	-0.213^* $[-0.29, -0.13]$	-0.201^* $[-0.28, -0.12]$	-0.197^* $[-0.27, -0.12]$	-0.204^* $[-0.28, -0.12]$	-0.170^* $[-0.24, -0.10]$
Urbanization (dist.)	-0.192^* $[-0.35, -0.08]$	-0.202^* $[-0.37, -0.09]$	-0.206^* $[-0.38, -0.09]$	-0.209^* $[-0.38, -0.09]$	-0.182^* $[-0.31, -0.11]$
Seniority (dist.)	-0.010^* $[-0.01, -0.01]$	-0.010^* $[-0.01, -0.01]$	-0.010^* $[-0.01, -0.01]$	-0.007^* $[-0.01, -0.00]$	-0.008^* $[-0.01, -0.00]$
Committee hom.	0.069^* $[0.04, 0.10]$	0.069^* $[0.04, 0.10]$	0.069^* $[0.04, 0.10]$	0.061^* $[0.04, 0.09]$	0.054^* $[0.03, 0.08]$

Table 18: **The full specification under alternative forms of the volume control.** Estimates of the terms that do not involve volume, from pilot fits of the chosen specification ($R = 100$; Section A.2). Columns: logged word volume (the main specification), with its centered square, as a three-degree-of-freedom natural spline, logged number of speeches instead of words, and no volume terms. 95% bootstrap confidence intervals in brackets; * indicates the interval excludes zero.

I.3 Equal-length networks

As a design-based complement to statistical control, we rebuilt the networks holding text quantity fixed. Within each Congress we set L to the 25th percentile of members' total one-minute word counts (252–549 words across Congresses), dropped members below L (about a quarter of members, by construction), sampled L words from each remaining member's concatenated speeches, and implemented the TF-IDF, cosine similarity, and row-wise z -score thresholding procedure. Table 19 reports the construction and diagnostics by Congress. In the equal-length networks, the in-degree-volume correlation decreases from 0.78 to 0.25, and in-degree correlates with the original networks at only $r \approx 0.29$. These results indicate that most of the raw in-degree variation is volume-driven, which aligns with the diagnostics above.

Congress	L (words)	Kept	Dropped	Density	In-deg. r vs. orig.	In-deg. $\times$ vol. (eq./orig.)
104	510	289	97	0.060	0.44	0.42 / 0.82
105	348	264	88	0.064	0.39	0.37 / 0.78
106	344	252	83	0.066	0.36	0.29 / 0.74
107	252	241	81	0.065	0.24	0.21 / 0.76
108	264	234	78	0.065	0.20	0.22 / 0.72
109	339	237	77	0.063	0.16	0.12 / 0.74
110	377	251	84	0.063	0.34	0.26 / 0.72
111	539	291	96	0.059	0.35	0.15 / 0.77
112	342	262	87	0.062	0.19	0.25 / 0.86
113	549	289	96	0.061	0.37	0.23 / 0.83
114	494	278	93	0.061	0.27	0.24 / 0.76
115	424	280	94	0.063	0.21	0.23 / 0.81
Mean				0.063	0.29	0.25 / 0.78

Table 19: **Equal-length network construction and diagnostics.** L is the per-Congress 25th percentile of members’ total one-minute word counts; members below L are dropped and all others are represented by exactly L sampled words. The final column reports the Spearman correlation of in-degree with speech count in the equal-length network and, after the slash, in the original network on the same members.

Table 20 reports the full specification re-estimated on the equal-length networks alongside the original estimates. The geometrically weighted in-degree coefficient is -3.04 $[-3.39, -2.34]$, suggesting that discursive concentration is sharper when every member contributes the same amount of text. The reciprocity coefficient also increases (5.08 $[4.85, 5.32]$), which is consistent with its origin in the symmetric similarity measure (Section H). Committee, urbanization, seniority and receiver-volume terms also retain their sign and statistical significance. Conversely, triadic closure falls from 1.11 to 0.16 and tie persistence from 0.48 to 0.18 , so the downsampling reinforces the concentration finding rather than the entirety of the endogenous terms.

The fine-grained dyadic ideology terms (absolute-NOMINATE and CF-score differences) are no longer statistically significant, which we attribute to the coarser lexical signal available in a few hundred sampled words per member. The partisan terms reorganize, with the same-party main effect flipping to be positive and significant (0.16), whereas its time interaction and the time trend are no longer significant. The sender-volume term flips to a small negative (-0.05), which is expected by construction. With text quantity standardized, the volume covariate no longer captures text-length and measures only residual speaking propensity. And three of the low-powered secondary terms that hover near the significance boundary across specifications (Section F) cross here as well, with chair homophily flipping to a small negative coefficient, and both minority-leader homophily and the minority-leader receiver effect turning positive. Note that the chair term that becomes statistically significant here is homophily, not the chair receiver effect noted in Section F, which changes sign under this construction. These terms are unstable in direction as well as in significance, and we decline to interpret them for this reason and because they do not back substantive claims in the paper. Instead, we treat the equal-length analysis as a robustness check on the concentration

finding specifically.

	Full (original networks)	Equal-length networks
Edges	−7.349 (−7.946;−6.715)*	−2.542 (−3.190;−2.064)*
Reciprocity	3.261 (3.034;3.477)*	5.076 (4.850;5.316)*
Triadic Closure	1.112 (1.026;1.184)*	0.160 (0.125;0.205)*
Open Triads	−0.232 (−0.250;−0.217)*	−0.251 (−0.267;−0.236)*
Tie Persistence ($t - 1$)	0.485 (0.418;0.546)*	0.178 (0.125;0.225)*
Linear Time Trend	−0.032 (−0.087;−0.001)*	0.007 (−0.036;0.034)
Party Hom.	−0.007 (−0.283;0.203)	0.162 (0.074;0.301)*
Party Hom. $\times$ Time	0.019 (0.004;0.049)*	−0.002 (−0.014;0.017)
DW-NOM 1 (Dist.)	−0.320 (−0.443;−0.185)*	0.003 (−0.147;0.097)
CF Score (Dist.)	−0.213 (−0.273;−0.141)*	0.021 (−0.051;0.106)
Speech Vol. (Pop.)	0.578 (0.531;0.629)*	0.137 (0.108;0.167)*
Speech Vol. (Soc.)	0.032 (0.008;0.063)*	−0.046 (−0.068;−0.017)*
Seniority (Dist.)	−0.010 (−0.013;−0.006)*	−0.009 (−0.011;−0.003)*
Urban. (Dist.)	−0.192 (−0.330;−0.056)*	−0.167 (−0.246;−0.077)*
Comm. Hom.	0.069 (0.043;0.097)*	0.056 (0.032;0.088)*
Gender Hom.	−0.014 (−0.052;0.019)	0.000 (−0.033;0.037)
Race Hom. (Black)	−0.013 (−0.075;0.062)	−0.024 (−0.139;0.073)
Eth. Hom. (Latino)	0.046 (−0.040;0.148)	0.044 (−0.016;0.096)
Maj. Leader Hom.	−0.038 (−0.102;0.021)	0.011 (−0.090;0.167)
Min. Leader Hom.	0.036 (−0.015;0.090)	0.145 (0.040;0.246)*
Comm. Chair Hom.	0.034 (−0.024;0.092)	−0.086 (−0.149;−0.003)*
Maj. Leader (Pop.)	0.103 (−0.060;0.284)	0.031 (−0.222;0.258)
Min. Leader (Pop.)	0.101 (−0.116;0.357)	0.206 (0.008;0.388)*
Comm. Chair (Pop.)	0.136 (−0.037;0.271)	−0.120 (−0.349;0.087)

* Null hypothesis value outside the confidence interval.

Table 20: **Equal-length-network refit.** Full specification estimated on networks rebuilt from equal-length member texts ($R = 100$), beside the original estimates. 95% bootstrap confidence intervals in parentheses. Because roughly the lowest-volume quarter of members is excluded by construction and each remaining member is represented by a few hundred sampled words, covariate estimates reflect a coarser lexical signal; the comparison targets the structural terms.

J Topic Communities: Detection and Labeling

As an additional descriptive foray into the speech network data, we consider the extent to which the parties focus on various topics. To do so, we employ the Louvain community detection algorithm to identify clusters of discussion topics. First, we apply the algorithm to detect clusters of members within the network. For each cluster, we calculate the average TF-IDF values of all words across its members. Next, we pass the terms with the highest scores to a large language model, Claude Opus 5.5 (Anthropic; model identifier `claude-opus-5-5`). On October 2, 2026, we provided each community’s 25 highest mean TF-IDF terms and instructed the model to assign each community a short, neutral topic label based on those 25 terms. The model was not given any other information about the communities, including their party composition. The session ran in Anthropic’s Claude Code interface, which does not expose sampling parameters such as temperature, so the default settings were applied. Because a model’s output can differ between runs, we treat the labels as data and include the prompt, term lists, and LLM-generated labels as part of the replication materials (available upon request). Labels were reviewed by the authors against the underlying term lists (Figure 7). Because members are assigned to one community, the labels capture each member’s most characteristic speech content rather than their full topical portfolio. Louvain partitions depend on the algorithm’s random starting order, so we fix the seed and report the resulting partition. Across 200 seeds for the 113th Congress, the partitions agree closely with the single realization reported here (median adjusted Rand index 0.84) and the climate community recurs in 188 out of the 200 runs (94%), where “recurrence” is defined as some community in the run having a Jaccard overlap of 0.5 or greater with the climate community in Figure 7.

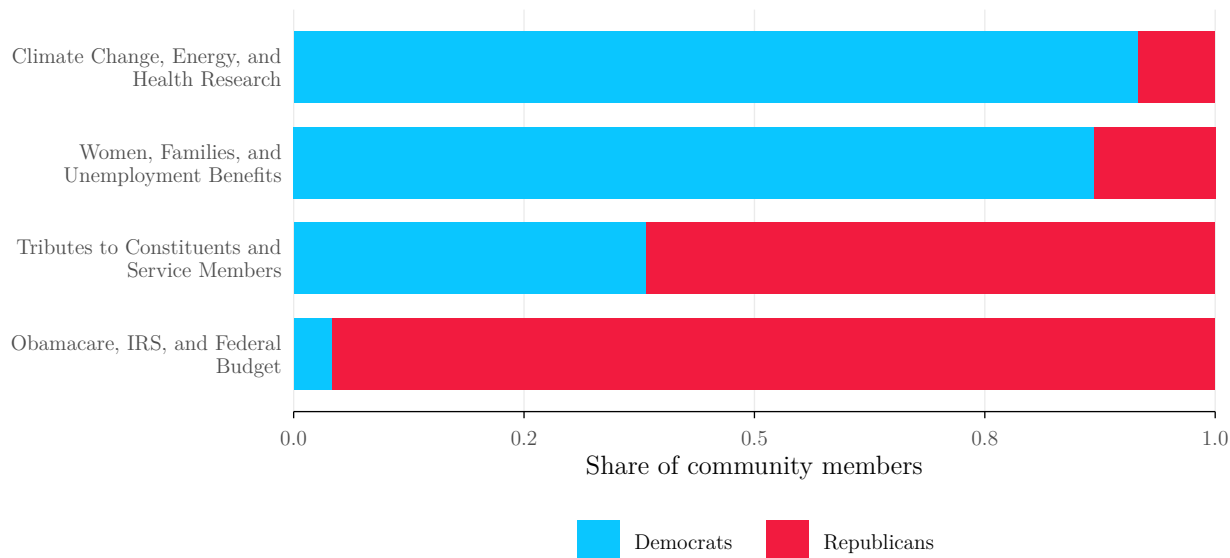


Figure 5: Proportion of Democratic (blue) and Republican (red) members of each Louvain topic community for the 113th Congress.

Figure 5 illustrates that these discussion communities align with the concept of issue

ownership ([Petrocik et al., 2003](#)), where Republicans and Democrats predominantly focus on issues traditionally associated with their respective parties. For example, the “climate change, energy, and health research” community is 92% Democratic, as expected, and the “women, families, and unemployment benefits” community is 87% Democratic. The “tributes to constituents and service members” community is the most bipartisan (38% Democratic). A notable exception is the community centered on health care (“Obamacare, IRS, and federal budget”), which is 96% Republican in this period, despite health care intuitively being viewed as an issue owned by the Democrats. However, this anomaly can be explained by the timing of the 113th Congress, which followed the passage of the Affordable Care Act. With the Democrats having achieved their policy objective, discussions on health care reform shifted to Republicans, particularly their opposition to the policy and myriad attempts to repeal it.

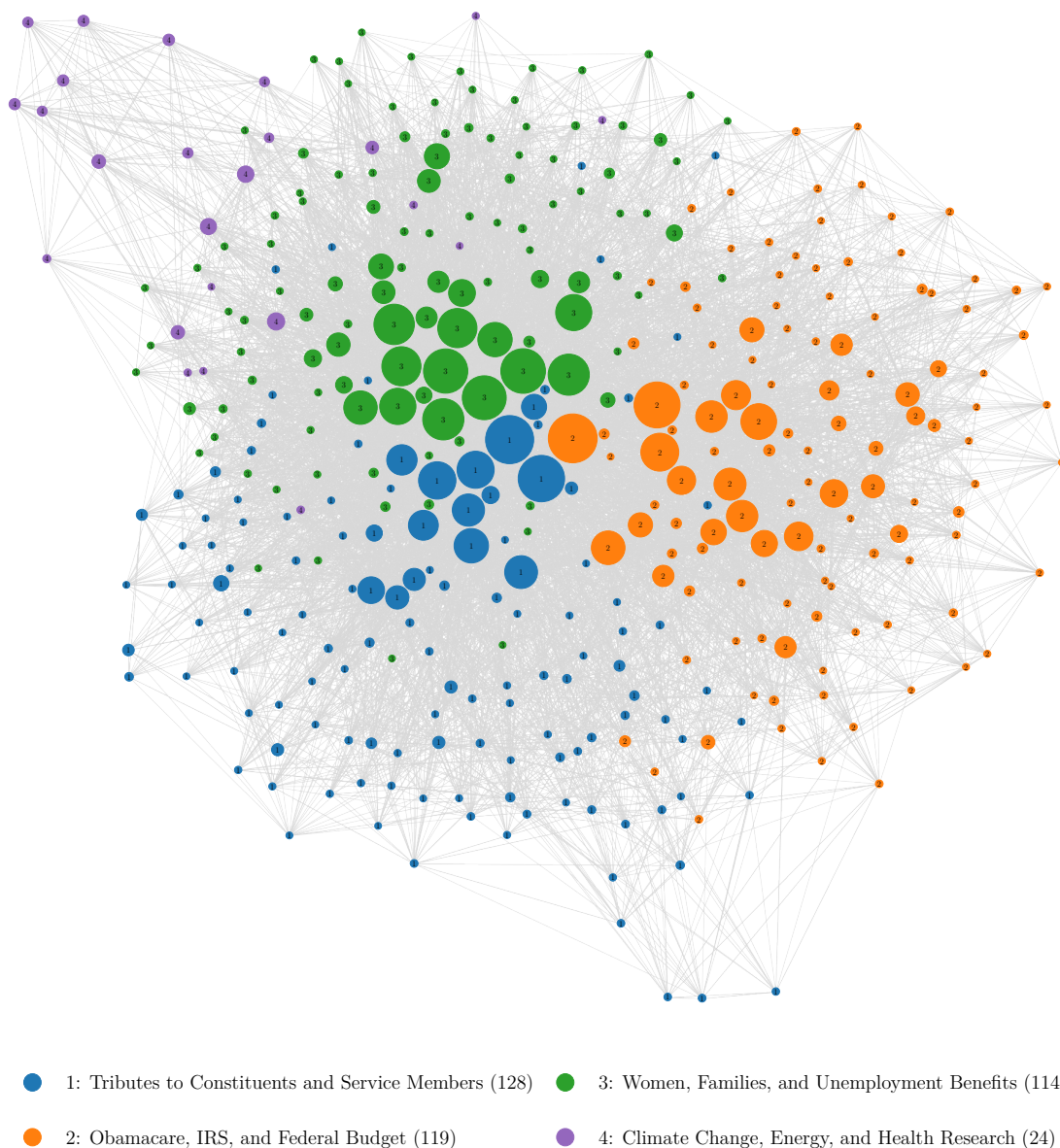


Figure 6: **Louvain topic communities for 113th Congress.** Node color indicates community membership and node size represents each member’s PageRank centrality. Communities were identified using the Louvain community detection algorithm and labeled from the most distinctive terms used by members within each community (labels drafted by a large language model and reviewed by the authors; see the text). Numbers on the nodes give each member’s community, and edges are drawn without arrowheads. Members are only able to be assigned to a single community, so while individual legislators may speak on multiple topics, only their most characteristic speech content is captured here. See Figure 7 for the highest weighted terms for each community. A spring layout positions more strongly connected members closer together and pushes unconnected members away, highlighting both the structure of speech-based communities and the thematic overlap between them.

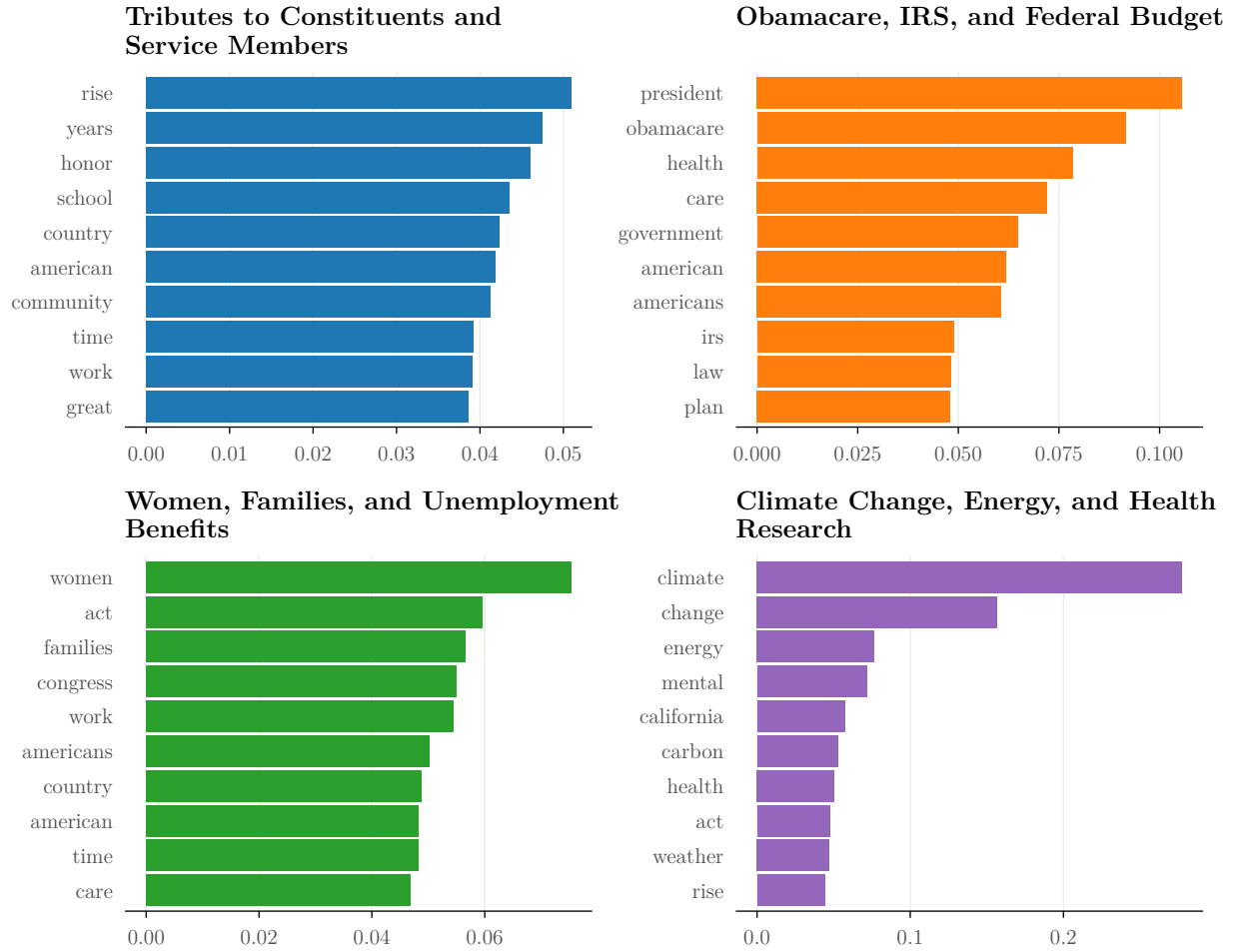


Figure 7: **Highest mean TF-IDF values for words by Louvain topic community for the 113th Congress.** Each panel displays the ten words with the highest mean TF-IDF values among members of a single Louvain-detected community. Bar length indicates relative term importance within the group. Colors correspond to the community color scheme used in the speech network visualization in Figure 6. Axes are scaled independently to emphasize within-group variation.

K Additional Descriptive Analysis

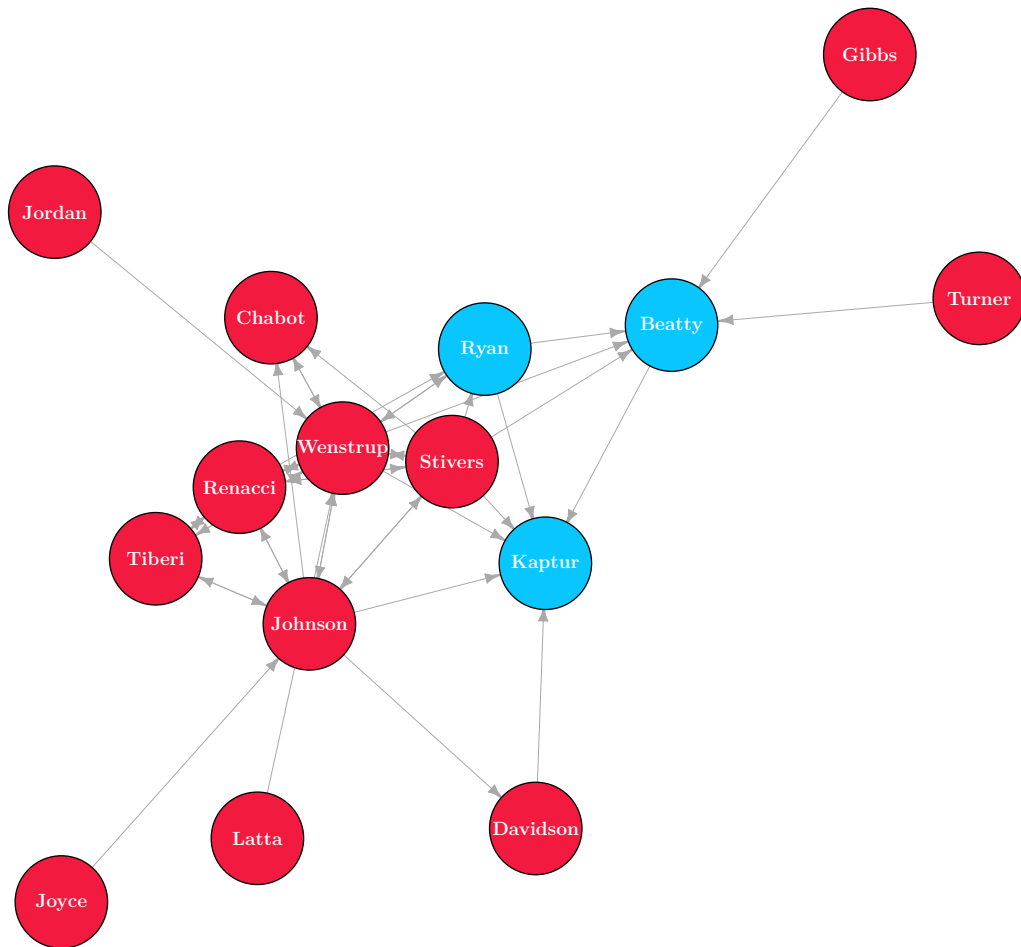


Figure 8: **One-minute speech network for congressional delegation from Ohio, 115th Congress.** This subgraph contains only representatives from Ohio to better illustrate individual connections than in the full network. Nodes are labeled with each member’s last name and colored by party affiliation (blue for Democrats, red for Republicans). Edges are directed: an edge $i \rightarrow j$ indicates that member j ’s language stands out within member i ’s distribution of lexical similarities. A spring layout is used to position nodes, with connected members placed closer together.

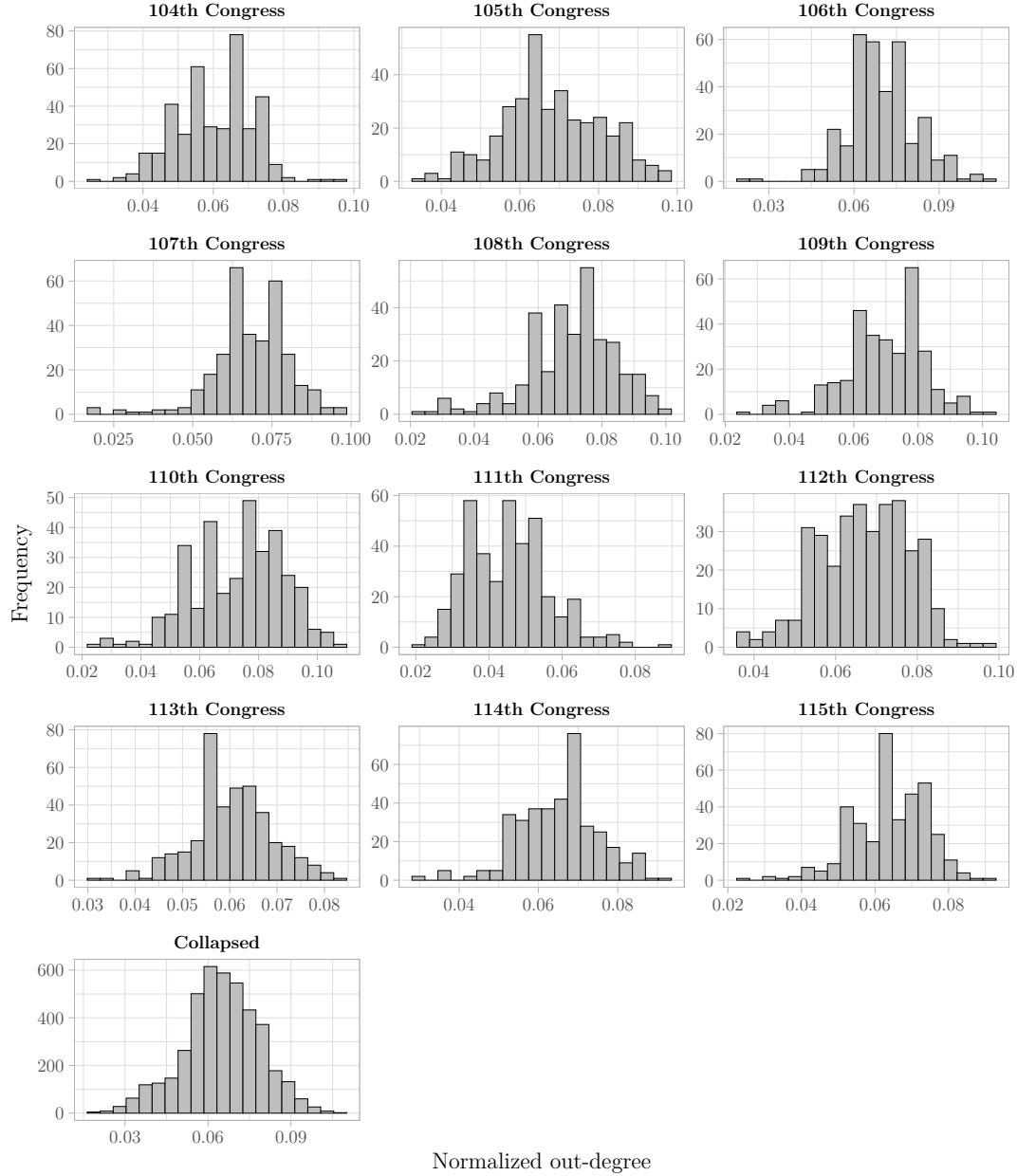


Figure 9: **Histograms of network out-degree centrality by session and collapsed.** Out-degree measures were normalized by dividing by $n - 1$, where n represents the number of nodes in each network.

The binarized network plots for each congressional session, shown in Figure 10, reveal two key insights about the underlying data. First, discourse frequently divides along party lines, with nodes forming distinct clusters that closely align with partisanship. Second, the node sizes—which scale with the square root of PageRank—indicate that within each congressional session, a small group of prominent members attracts a disproportionate share of the chamber’s discursive attention, while most members remain on the periphery of the central discussion.

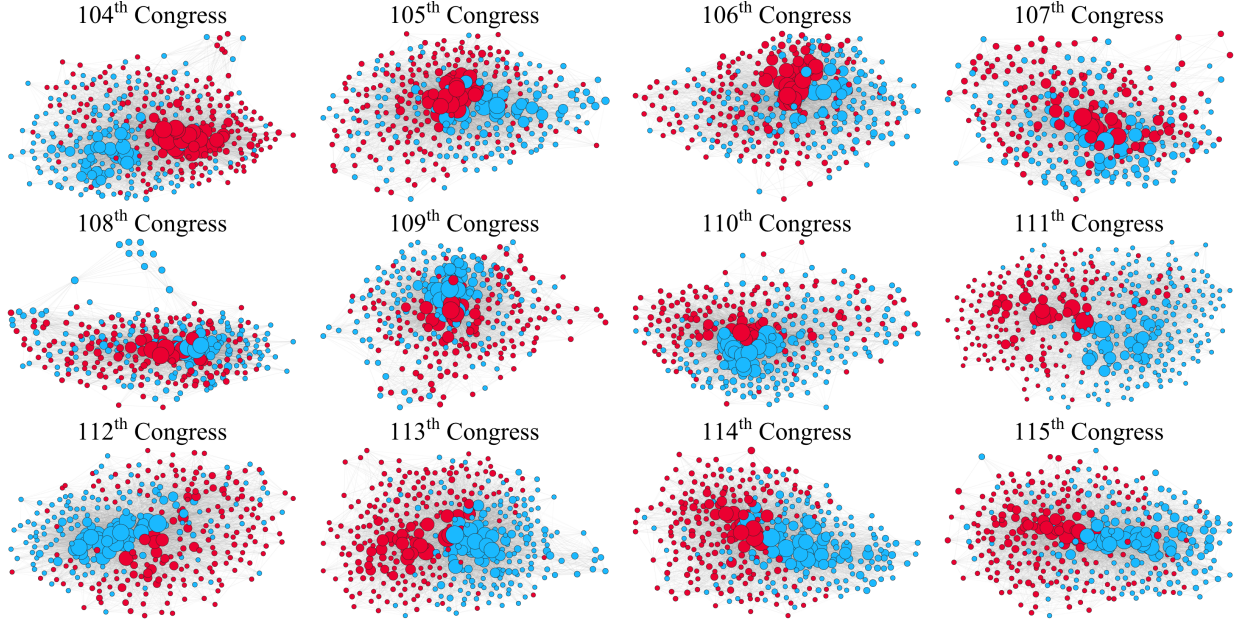


Figure 10: **Congressional speech networks by session, 104th–115th Congresses.** Node color indicates political party; nodes are colored blue to indicate a member of the Democratic Party, and red otherwise. Node size scales with the square root of each member’s PageRank centrality (normalized within panel) and edges are drawn translucently.

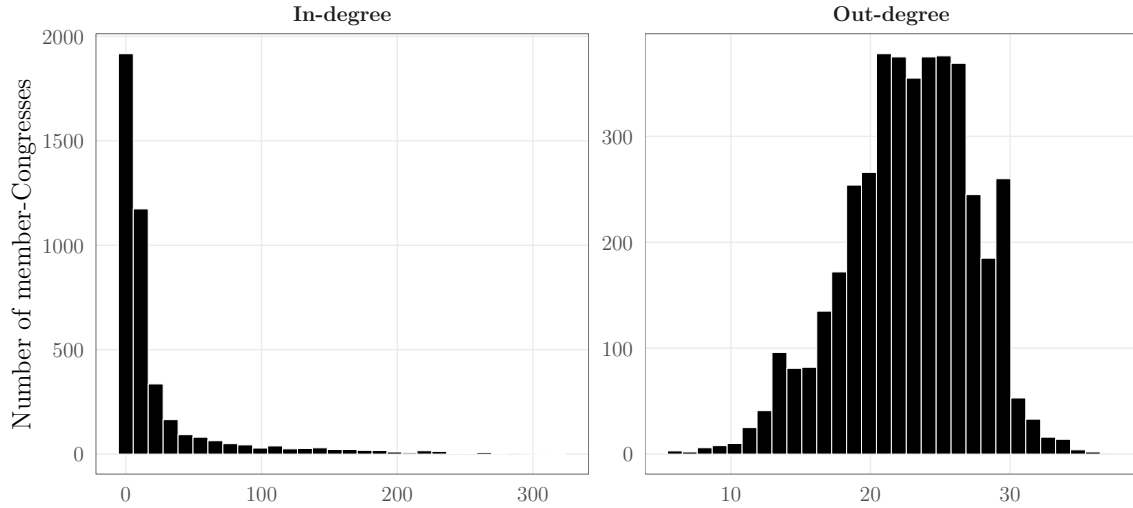


Figure 11: **In- and out-degree distribution across all Congresses.**

Social networks often exhibit a power-law distribution, where a few nodes are highly connected and influential, while the majority have few connections ([Barabási and Albert, 1999](#); [Barabási et al., 2002](#)). Grounded in existing scholarship, we expect one-minute speech networks to reflect a similarly concentrated distribution of discursive attention.

The directionality of the ties in the network follows from the row-wise construction of

the measure. An edge $i \rightarrow j$ indicates that member j 's language stands out within member i 's own distribution of lexical similarities. Therefore, a member's in-degree represents the number of colleagues for whom that member is an unusually strong lexical match, relative to each colleague's own baseline. In-degree captures how widely a member's language stands out across the chamber. Because cosine similarity is symmetric, direction dictates whose baseline a similarity is compared with, rather than who influences whom. Conversely, out-degree captures how many distinct standouts a member's own speech aligns with. As depicted in Figure 11, the in-degree distribution within these one-minute speech networks is markedly skewed, suggesting a pattern in which only a select few members' language stands out to many colleagues, while most members' speech stands out to few. In Table 21 we show the best-connected members as determined by in-degree centrality in the network. From the 104th to the 115th Congress the members with the highest in-degree are Martin Hoke (OH-10), Sheila Jackson Lee (TX-18), Andrea Seastrand (CA-22), Janice Hahn (CA-36, 44), and Doug LaMalfa (CA-1).

Table 21: **Top 10 In-Degree Centrality Members, Collapsed.** Values rounded to three decimal places. Collapsed values represent the mean centrality score for each member across all sessions they served in from the 104th to 115th Congresses.

House Representative	State	District	Party	In-Degree Centrality
Martin Hoke	OH	10	R	0.740
Sheila Jackson Lee	TX	18	D	0.655
Andrea Seastrand	CA	22	R	0.592
Janice Hahn	CA	36, 44	D	0.556
Doug LaMalfa	CA	1	R	0.554
Virginia Foxx	NC	5	R	0.545
Kevin Yoder	KS	3	R	0.544
Joseph Pitts	PA	16	R	0.542
Robert Dold	IL	10	R	0.538
David Cicilline	RI	1	D	0.496

An important property of this measure is its relationship to speaking propensity. Members differ in how often they take the floor, ranging from one to 212 one-minute speeches per Congress. A member who speaks on more occasions covers more lexical ground, mechanically increasing the chance of standing out in colleagues' similarity distributions. We therefore control directly for members' speech volume in the inferential models above.

We further explore the congressional speech networks by visualizing lexical similarity within and across parties over time. Figure 12 displays the trend in cosine similarity for each sender–receiver party pair. All four panels show a decline between the 104th and 107th Congresses. Similarity among members of the same party, particularly Democrats, rises during and just after the Democratic majority of the 110th and 111th Congresses. Notably, similarity for Republican senders, toward both Republicans and Democrats, spikes in the 111th Congress and falls back immediately afterward. This sharp increase may reflect coordinated minority-party messaging in the last Congress before Republicans regained the

majority. Figure 13 reports the corresponding tie densities by sender and receiver party.

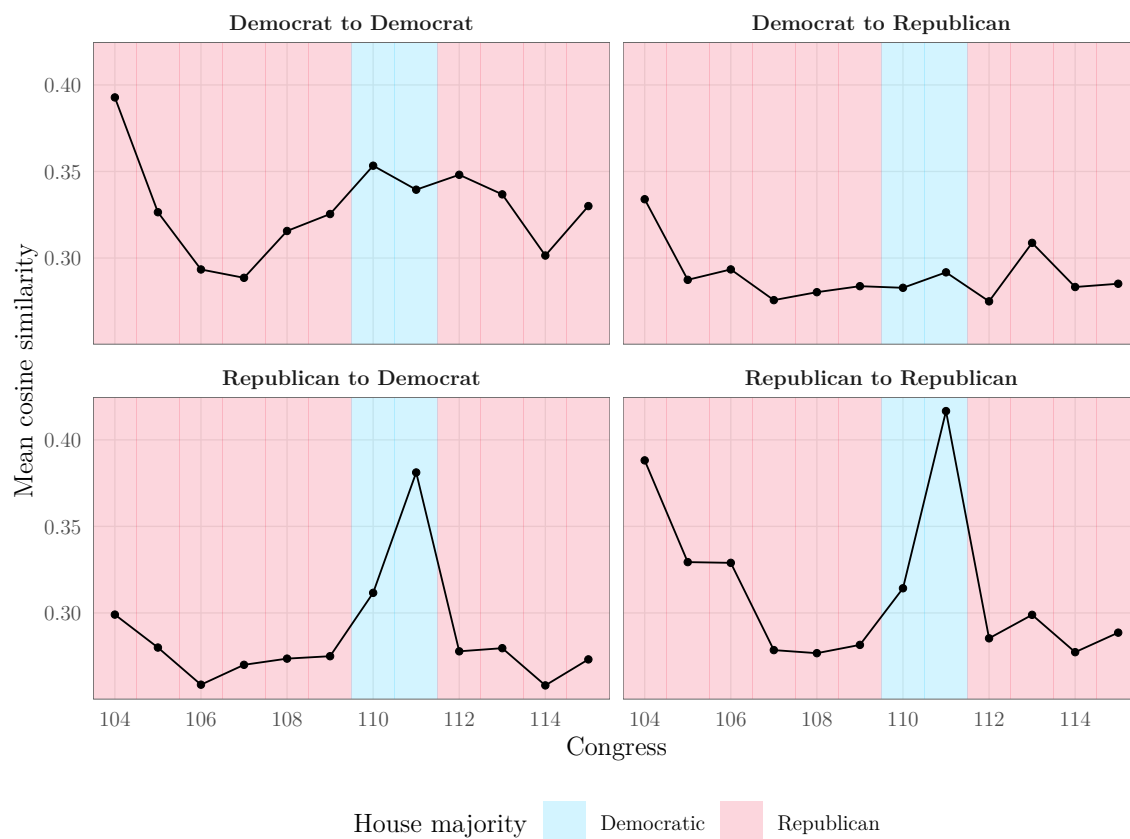


Figure 12: Mean cosine similarity by partisanship, 104th – 115th Congresses. Shading represents majority party.

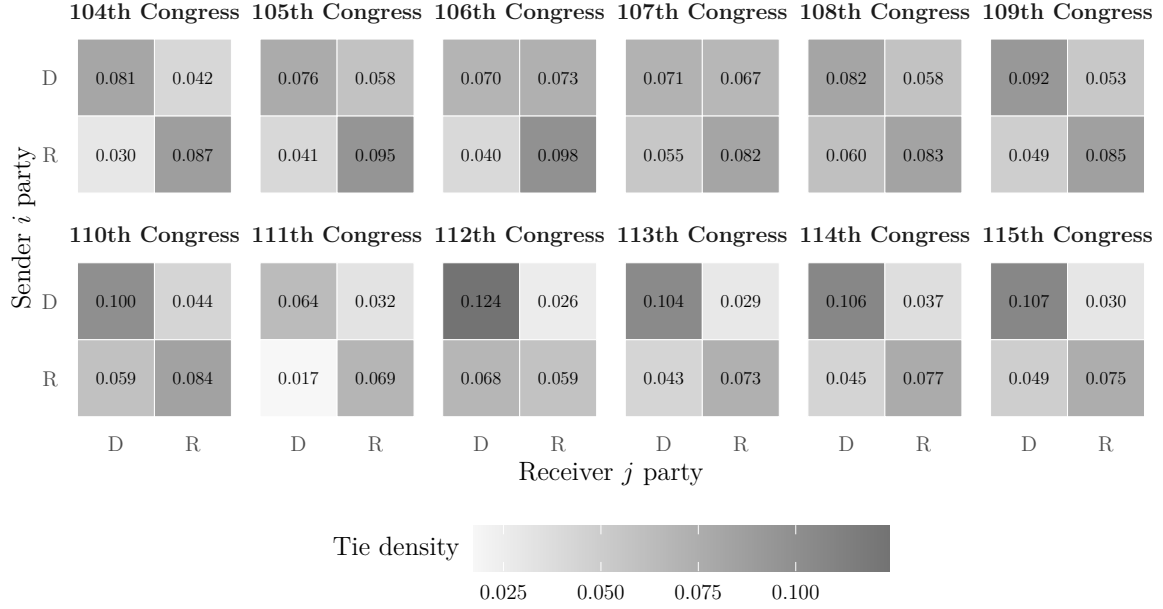


Figure 13: **Tie density by sender and receiver party, 104th–115th Congresses.** Each cell displays the observed share out of all possible directed ties from senders i of one party (rows) to receivers j of the same or the other party (columns); same-party blocks exclude self-ties, and darker shading indicates higher density. The figure extends the block densities of Figure 1 in the main text to all modeled Congresses. Averaging the two parties, same-party blocks are denser than cross-party blocks in all Congresses, by a factor ranging from 1.25 (107th) to 2.65 (111th). In two Congresses, one party’s members send ties across the aisle slightly more often than within it: Democrats in the 106th (0.073 versus 0.070) and Republicans in the 112th (0.068 versus 0.059). The single independent member in each of the 104th–107th Congresses is omitted.

L Centrality and Summary Tables

Table 22: **Top Ten In-Degree Centrality Members, 104th to 115th Congresses and Collapsed.** Values rounded to three decimal places. Collapsed values represent the mean centrality score for each member across all sessions they served in from the 104th to 115th Congresses.

Congress	Name	State	District	Party	In-Degree Centrality
Collapsed	Martin Hoke	OH	10	R	0.740
	Sheila Jackson Lee	TX	18	D	0.655
	Andrea Seastrand	CA	22	R	0.592
	Janice Hahn	CA	36, 44	D	0.556
	Doug LaMalfa	CA	1	R	0.554
	Virginia Foxx	NC	5	R	0.545
	Kevin Yoder	KS	3	R	0.544
	Joseph Pitts	PA	16	R	0.542
	Robert Dold	IL	10	R	0.538
	David Cicilline	RI	1	D	0.496
104	Martin Hoke	OH	10	R	0.740
	Todd Tiahrt	KS	4	R	0.670
	Clifford Stearns	FL	6	R	0.608
	Jack Kingston	GA	1	R	0.597
	Andrea Seastrand	CA	22	R	0.592
	Cass Ballenger	NC	10	R	0.566
	Joel Hefley	CO	5	R	0.556
	Walter Jones	NC	3	R	0.522
	Sheila Jackson Lee	TX	18	D	0.514
	Steve Chabot	OH	1	R	0.504

105	James Gibbons	NV	2	R	0.823
	Sheila Jackson Lee	TX	18	D	0.738
	Gilbert Gutknecht	MN	1	R	0.672
	Jack Kingston	GA	1	R	0.670
	Rosa DeLauro	CT	3	D	0.613
	James McGovern	MA	3, 2	D	0.607
	Steve Chabot	OH	1	R	0.575
	Todd Tiahrt	KS	4	R	0.538
	Joseph Pitts	PA	16	R	0.521
	Cass Ballenger	NC	10	R	0.516
106	Joseph Pitts	PA	16	R	0.913
	James Gibbons	NV	2	R	0.781
	Cass Ballenger	NC	10	R	0.644
	Clifford Stearns	FL	6	R	0.590
	Roscoe Bartlett	MD	6	R	0.572
	Sheila Jackson Lee	TX	18	D	0.554
	Robert Schaffer	CO	4	R	0.551
	Thomas Tancredo	CO	6	R	0.518
	Jack Kingston	GA	1	R	0.515
	Raymond Green	TX	29	D	0.491
107	Mike Pence	IN	2, 6	R	0.745
	Sheila Jackson Lee	TX	18	D	0.729
	James Gibbons	NV	2	R	0.720
	Joseph Pitts	PA	16	R	0.707
	Mark Foley	FL	16	R	0.692
	Loretta Sanchez	CA	46, 47	D	0.583
	Hilda Solis	CA	31, 32	D	0.458
	Sam Johnson	TX	3	R	0.421
	Clifford Stearns	FL	6	R	0.411
	Jim Ryun	KS	2	R	0.389

108	Loretta Sanchez	CA	46, 47	D	0.752
	Sheila Jackson Lee	TX	18	D	0.714
	Joseph Pitts	PA	16	R	0.698
	Mike Pence	IN	2, 6	R	0.621
	Thomas Delay	TX	22	R	0.598
	Michael Burgess	TX	26	R	0.595
	James McDermott	WA	7	D	0.588
	Marsha Blackburn	TN	7	R	0.582
	Mark Foley	FL	16	R	0.540
	Rahm Emanuel	IL	5	D	0.537
109	Virginia Foxx	NC	5	R	0.796
	Sheila Jackson Lee	TX	18	D	0.760
	Tom Price	GA	6	R	0.700
	Marsha Blackburn	TN	7	R	0.623
	Joseph Pitts	PA	16	R	0.601
	Earl Blumenauer	OR	3	D	0.594
	Phil Gingrey	GA	11	R	0.572
	Mike Pence	IN	2, 6	R	0.569
	Hilda Solis	CA	31, 32	D	0.546
	Rahm Emanuel	IL	5	D	0.502
110	Stephen Cohen	TN	9	D	0.796
	Virginia Foxx	NC	5	R	0.781
	Joseph Pitts	PA	16	R	0.689
	Marsha Blackburn	TN	7	R	0.668
	Bruce Braley	IA	1	D	0.587
	Sheila Jackson Lee	TX	18	D	0.548
	John Yarmuth	KY	3	D	0.539
	Russ Carnahan	MO	3	D	0.512
	Tim Walz	MN	1	D	0.497
	Frank, Pallone	NJ	6	D	0.494

111	John Fleming	LA	4	R	0.728
	John Yarmuth	KY	3	D	0.694
	Joseph Pitts	PA	16	R	0.536
	Sheila Jackson Lee	TX	18	D	0.523
	Donna Edwards	MD	4	D	0.453
	John Boehner	OH	8	R	0.446
	Virginia Foxx	NC	5	R	0.438
	Diane Watson	CA	32, 33	D	0.399
	Mike Pence	IN	2, 6	R	0.376
	Paul Tonko	NY	21, 20	D	0.370
	Sam Johnson	TX	3	R	0.370
112	Sheila Jackson Lee	TX	18	D	0.784
	David Cicilline	RI	1	D	0.782
	Stephen Cohen	TN	9	D	0.759
	Tim Walz	MN	1	D	0.641
	John Yarmuth	KY	3	D	0.632
	Albio Sires	NJ	13, 8	D	0.483
	Janice Hahn	CA	36, 44	D	0.471
	Judy Chu	CA	32, 27	D	0.468
	Kathleen Hochul	NY	26	D	0.457
	Lois Capps	CA	22, 23, 24	D	0.434
113	Sheila Jackson Lee	TX	18	D	0.831
	Glenn Thompson	PA	5	R	0.698
	Virginia Foxx	NC	5	R	0.612
	Kevin Yoder	KS	3	R	0.586
	Alice Titus	NV	3, 1	D	0.576
	Janice Hahn	CA	36, 44	D	0.570
	David Cicilline	RI	1	D	0.531
	Barbara Lee	CA	9, 13	D	0.443
	Luke Messer	IN	6	R	0.438
	Susan Davis	CA	49, 53	D	0.430

114	Doug LaMalfa	CA	1	R	0.708
	Erik Paulsen	MN	3	R	0.670
	Robert Dold	IL	10	R	0.670
	Kevin Yoder	KS	3	R	0.659
	Janice Hahn	CA	36, 44	D	0.627
	Loretta Sanchez	CA	46, 47	D	0.619
	Glenn Thompson	PA	5	R	0.597
	Virginia Foxx	NC	5	R	0.505
	Ileana Ros-Lehtinen	FL	18, 27	R	0.489
	Steny Hoyer	MD	5	D	0.476
115	Sheila Jackson Lee	TX	18	D	0.764
	Doug LaMalfa	CA	1	R	0.761
	Brad Schneider	IL	10	D	0.590
	Tulsi Gabbard	HI	2	D	0.576
	Keith Rothfus	PA	12	R	0.555
	Kevin Yoder	KS	3	R	0.547
	Donald, Payne	NJ	10	D	0.534
	Glenn Thompson	PA	5	R	0.491
	Jared Polis	CO	2	D	0.461
	Raja Krishnamoorthi	IL	8	D	0.456

Table 23: **Top Ten Betweenness Centrality Members, 104th to 115th Congresses and Collapsed.** Values rounded to three decimal places. Collapsed values represent the mean centrality score for each member across all sessions they served in from the 104th to 115th Congresses.

Congress	Name	State	District	Party	Betweenness Centrality
Collapsed	Gillespie Montgomery	MS	3	D	0.179
	David Funderburk	NC	2	R	0.161
	Robert Dornan	CA	46	R	0.147
	Gabrielle Giffords	AZ	8	D	0.099
	Ann Kirkpatrick	AZ	1	D	0.099
	Ed Bryant	TN	7	R	0.063
	Max Burns	GA	12	R	0.055
	Kathleen Hochul	NY	26	D	0.053
	Ted Poe	TX	2	R	0.052
	Ed Pastor	AZ	2, 4	D	0.051
104	Gillespie Montgomery	MS	3	D	0.179
	David Funderburk	NC	2	R	0.161
	Robert Dornan	CA	46	R	0.147
	Julius Watts	OK	4	R	0.143
	Isaac Skelton	MO	4	D	0.142
	Lamar Smith	TX	21	R	0.139
	Ed Bryant	TN	7	R	0.123
	Julian Dixon	CA	32	D	0.091
	William Thornberry	TX	13	R	0.067
	Daniel Schaefer	CO	6	R	0.067

105	Bobby Etheridge	NC	2	D	0.072
	Ileana Ros-Lehtinen	FL	18, 27	R	0.063
	Charles Stenholm	TX	17	D	0.057
	Raymond Green	TX	29	D	0.052
	Douglas Bereuter	NE	1	R	0.047
	William Barrett	NE	3	R	0.046
	Michael McNulty	NY	21	D	0.044
	Calvin Dooley	CA	20	D	0.043
	James McGovern	MA	3, 2	D	0.039
	Joseph Pitts	PA	16	R	0.038
106	Sheila Jackson Lee	TX	18	D	0.087
	Dennis Kucinich	OH	10	D	0.061
	Ileana Ros-Lehtinen	FL	18, 27	R	0.059
	Sanford Bishop	GA	2	D	0.052
	Diana DeGette	CO	1	D	0.044
	John Duncan	TN	2	R	0.043
	Stephanie Jones	OH	11	D	0.039
	Joseph Pitts	PA	16	R	0.033
	Nicholas Lampson	TX	9, 22	D	0.032
	Thomas Davis	VA	11	R	0.032
107	Frank Wolf	VA	10	R	0.054
	Loretta Sanchez	CA	46, 47	D	0.054
	Ileana Ros-Lehtinen	FL	18, 27	R	0.044
	Addison Wilson	SC	2	R	0.037
	Mike Pence	IN	2, 6	R	0.035
	David Dreier	CA	28, 26	R	0.023
	Constance Morella	MD	8	R	0.023
	Lois Capps	CA	22, 23, 24	D	0.023
	Nicholas Lampson	TX	9, 22	D	0.021
	Daniel Miller	FL	13	R	0.020

108	Max Burns	GA	12	R	0.055
	Michael Collins	GA	3, 8	R	0.055
	Denny Rehberg	MT	1	R	0.038
	J. Forbes	VA	4	R	0.037
	John Boozman	AR	3	R	0.035
	Ileana Ros-Lehtinen	FL	18, 27	R	0.034
	Randy Neugebauer	TX	19	R	0.032
	Nicholas Lampson	TX	9, 22	D	0.032
	Mark Kennedy	MN	2, 6	R	0.031
	Loretta Sanchez	CA	46, 47	D	0.030
109	K. Conaway	TX	11	R	0.095
	Steve Chabot	OH	1	R	0.078
	Ted Poe	TX	2	R	0.072
	Lamar Smith	TX	21	R	0.050
	Zoe Lofgren	CA	16, 19	D	0.049
	John Boozman	AR	3	R	0.049
	Rush Holt	NJ	12	D	0.043
	Michael Burgess	TX	26	R	0.043
	Paul Ryan	WI	1	R	0.042
	Janice Schakowsky	IL	9	D	0.040
110	John Boozman	AR	3	R	0.131
	Phil Gingrey	GA	11	R	0.101
	John Dingell	MI	16, 15	D	0.067
	Allyson Schwartz	PA	13	D	0.065
	Stephen Cohen	TN	9	D	0.058
	Charles Boustany	LA	7, 3	R	0.055
	Nicola Tsongas	MA	5, 3	D	0.039
	Thelma Drake	VA	2	R	0.039
	Paul Ryan	WI	1	R	0.039
	Patrick McHenry	NC	10	R	0.038

111	Ann Kirkpatrick	AZ	1	D	0.195
	Ed Pastor	AZ	2, 4	D	0.192
	Gabrielle Giffords	AZ	8	D	0.192
	Ted Poe	TX	2	R	0.188
	Stephanie H. Sandlin	SD	1	D	0.175
	Joseph Pitts	PA	16	R	0.123
	Betty Sutton	OH	13	D	0.068
	Marcia Fudge	OH	11	D	0.065
	Lois Capps	CA	22, 23, 24	D	0.051
	Jay Inslee	WA	1	D	0.045
112	Gregory Meeks	NY	6, 5	D	0.078
	Brian Higgins	NY	27, 26	D	0.076
	Kathleen Hochul	NY	26	D	0.053
	Michael Fitzpatrick	PA	8	R	0.047
	David Cicilline	RI	1	D	0.047
	Glenn Thompson	PA	5	R	0.046
	Mike Coffman	CO	6	R	0.043
	Jaime Herrera Beutler	WA	3	R	0.041
	Judy Biggert	IL	13	R	0.038
	Erik Paulsen	MN	3	R	0.038
113	Virginia Foxx	NC	5	R	0.053
	Jason Smith	MO	8	R	0.045
	Tim Griffin	AR	2	R	0.041
	Bob Gibbs	OH	18, 7	R	0.038
	Marc Veasey	TX	33	D	0.037
	Richard Hudson	NC	8	R	0.034
	Kenny Marchant	TX	24	R	0.034
	Marlin Stutzman	IN	3	R	0.031
	Glenn Thompson	PA	5	R	0.030
	Lamar Smith	TX	21	R	0.030

114	Sheila Jackson Lee	TX	18	D	0.048
	Robert Dold	IL	10	R	0.045
	Rodney Davis	IL	13	R	0.044
	Ted Poe	TX	2	R	0.034
	Bradley Walker	NC	6	R	0.032
	Dan Kildee	MI	5	D	0.032
	Ileana Ros-Lehtinen	FL	18, 27	R	0.030
	Marcia Kaptur	OH	9	D	0.029
	Joyce Beatty	OH	3	D	0.029
	Theodore Deutch	FL	19, 21, 22	D	0.027
115	Raymond Green	TX	29	D	0.057
	Rodney Davis	IL	13	R	0.050
	Erik Paulsen	MN	3	R	0.046
	Claudia Tenney	NY	22	R	0.044
	Virginia Foxx	NC	5	R	0.039
	Mike Coffman	CO	6	R	0.039
	Marcia Kaptur	OH	9	D	0.037
	Susan Brooks	IN	5	R	0.036
	Ileana Ros-Lehtinen	FL	18, 27	R	0.031
	John Shimkus	IL	20, 19, 15	R	0.029

Table 24: **Global Network Summary Statistics.** Values rounded to three decimal places. Collapsed network measures are calculated over all congress sessions analyzed. Under collapsed, total node and edge counts represent the sum over all congress sessions, and mean values represent the sum of the measures for each congress divided by the number of congress sessions (twelve).

Congress	Network Measure	Value
Collapsed	Total Node Count	4222
	Total Edge Count	96126
	Mean Density	0.065
	Mean Global Clustering Coefficient	0.291
	Mean Average Path Length	3.769
104	Node Count	386
	Edge Count	8965
	Density	0.06
	Global Clustering Coefficient	0.279
	Average Path Length	5.24
105	Node Count	352
	Edge Count	8434
	Density	0.068
	Global Clustering Coefficient	0.296
	Average Path Length	3.867
106	Node Count	335
	Edge Count	7854
	Density	0.07
	Global Clustering Coefficient	0.304
	Average Path Length	3.385
107	Node Count	322
	Edge Count	7107
	Density	0.069
	Global Clustering Coefficient	0.297
	Average Path Length	3.007

108	Node Count	312
	Edge Count	6874
	Density	0.071
	Global Clustering Coefficient	0.309
	Average Path Length	3.41
109	Node Count	314
	Edge Count	6881
	Density	0.07
	Global Clustering Coefficient	0.305
	Average Path Length	3.732
110	Node Count	335
	Edge Count	8174
	Density	0.073
	Global Clustering Coefficient	0.328
	Average Path Length	3.798
111	Node Count	387
	Edge Count	6793
	Density	0.045
	Global Clustering Coefficient	0.236
	Average Path Length	5.019
112	Node Count	349
	Edge Count	8131
	Density	0.067
	Global Clustering Coefficient	0.293
	Average Path Length	3.722
113	Node Count	385
	Edge Count	9008
	Density	0.061
	Global Clustering Coefficient	0.271
	Average Path Length	3.427

114	Node Count	371
	Edge Count	8953
	Density	0.065
	Global Clustering Coefficient	0.282
	Average Path Length	3.192
115	Node Count	374
	Edge Count	8952
	Density	0.064
	Global Clustering Coefficient	0.288
	Average Path Length	3.429

Table 25: **Local Network Summary Statistics.** Values rounded to three decimal places. Cells marked with “*” represent nonzero values smaller than 0.001. Values of each measure are first calculated for each member of Congress, and the corresponding summary statistics are presented for each congress and collapsed over all congresses.

Congress	Network Measure	Mean	SD	Min.	1st Qu.	Median	3rd Qu.	Max.
Collapsed	Betweenness Centrality	0.007	0.013	0	*	0.003	0.008	0.195
	Eigenvector Centrality	0.019	0.05	0	*	*	0.004	0.339
	In-Degree	0.065	0.123	0	0.006	0.019	0.054	0.913
	Local Clustering Coefficient	0.534	0.16	0.09	0.429	0.551	0.649	1
	Out-Degree	0.065	0.014	0.019	0.057	0.065	0.075	0.108
	PageRank	0.003	0.006	*	*	*	0.001	0.049
	Total Degree	0.13	0.127	0.028	0.069	0.086	0.123	0.991
104	Betweenness Centrality	0.009	0.023	0	*	0.001	0.008	0.179
	Eigenvector Centrality	0.018	0.048	0	0	*	*	0.236
	In-Degree	0.06	0.124	0	0.005	0.013	0.036	0.74
	Local Clustering Coefficient	0.547	0.183	0.139	0.419	0.558	0.682	0.971
	Out-Degree	0.06	0.011	0.026	0.052	0.061	0.068	0.096
	PageRank	0.003	0.006	*	*	*	*	0.029
	Total Degree	0.121	0.127	0.034	0.062	0.075	0.101	0.81
105	Betweenness Centrality	0.007	0.011	0	*	0.002	0.007	0.072
	Eigenvector Centrality	0.019	0.05	0	*	*	*	0.219
	In-Degree	0.068	0.133	0	0.006	0.017	0.055	0.823
	Local Clustering Coefficient	0.535	0.171	0.09	0.403	0.547	0.662	0.915
	Out-Degree	0.068	0.013	0.034	0.06	0.068	0.077	0.097
	PageRank	0.003	0.006	*	*	*	0.001	0.028
	Total Degree	0.137	0.138	0.037	0.071	0.088	0.12	0.912
106	Betweenness Centrality	0.007	0.01	0	*	0.003	0.009	0.087
	Eigenvector Centrality	0.019	0.051	0	*	*	0.001	0.226
	In-Degree	0.07	0.126	0	0.009	0.024	0.061	0.913
	Local Clustering Coefficient	0.515	0.157	0.128	0.399	0.513	0.623	0.894
	Out-Degree	0.07	0.012	0.021	0.063	0.069	0.078	0.108
	PageRank	0.003	0.006	*	*	*	0.001	0.032
	Total Degree	0.14	0.13	0.039	0.078	0.093	0.139	0.991
107	Betweenness Centrality	0.006	0.007	0	0.001	0.004	0.008	0.054
	Eigenvector Centrality	0.021	0.052	0	*	0.001	0.01	0.269
	In-Degree	0.069	0.115	0	0.013	0.031	0.062	0.745
	Local Clustering Coefficient	0.513	0.142	0.14	0.414	0.527	0.618	0.933
	Out-Degree	0.069	0.012	0.019	0.062	0.068	0.078	0.097
	PageRank	0.003	0.006	*	*	*	0.002	0.033
	Total Degree	0.138	0.117	0.028	0.081	0.103	0.139	0.816

108	Betweenness Centrality	0.007	0.009	0	0.001	0.004	0.01	0.056
	Eigenvector Centrality	0.02	0.053	0	*	*	0.004	0.233
	In-Degree	0.071	0.131	0	0.013	0.026	0.061	0.752
	Local Clustering Coefficient	0.565	0.173	0.153	0.445	0.574	0.7	1
	Out-Degree	0.071	0.014	0.022	0.064	0.071	0.08	0.1
	PageRank	0.003	0.007	*	*	*	0.002	0.032
	Total Degree	0.142	0.134	0.032	0.08	0.096	0.135	0.83
109	Betweenness Centrality	0.008	0.012	0	*	0.004	0.01	0.095
	Eigenvector Centrality	0.02	0.053	0	*	*	0.002	0.245
	In-Degree	0.07	0.131	0	0.01	0.022	0.051	0.796
	Local Clustering Coefficient	0.563	0.162	0.11	0.457	0.596	0.676	0.871
	Out-Degree	0.07	0.013	0.026	0.064	0.07	0.08	0.102
	PageRank	0.003	0.006	*	*	*	0.001	0.033
	Total Degree	0.14	0.135	0.042	0.077	0.093	0.125	0.879
110	Betweenness Centrality	0.008	0.013	0	*	0.004	0.009	0.131
	Eigenvector Centrality	0.019	0.051	0	*	*	*	0.207
	In-Degree	0.073	0.135	0	0.009	0.021	0.052	0.796
	Local Clustering Coefficient	0.567	0.168	0.153	0.444	0.584	0.703	0.874
	Out-Degree	0.073	0.015	0.024	0.063	0.075	0.084	0.108
	PageRank	0.003	0.006	*	*	*	0.001	0.028
	Total Degree	0.146	0.141	0.033	0.078	0.093	0.132	0.895
111	Betweenness Centrality	0.008	0.024	0	*	*	0.005	0.195
	Eigenvector Centrality	0.016	0.048	0	0	*	0.001	0.339
	In-Degree	0.046	0.095	0	0.003	0.01	0.038	0.728
	Local Clustering Coefficient	0.483	0.145	0.097	0.393	0.496	0.586	0.802
	Out-Degree	0.046	0.011	0.021	0.036	0.044	0.053	0.088
	PageRank	0.003	0.007	*	*	*	*	0.049
	Total Degree	0.091	0.094	0.028	0.049	0.062	0.086	0.759
112	Betweenness Centrality	0.006	0.011	0	*	0.002	0.006	0.078
	Eigenvector Centrality	0.018	0.05	0	*	*	0.002	0.231
	In-Degree	0.067	0.128	0	0.006	0.014	0.052	0.784
	Local Clustering Coefficient	0.533	0.155	0.134	0.445	0.562	0.644	0.868
	Out-Degree	0.067	0.011	0.037	0.06	0.066	0.075	0.098
	PageRank	0.003	0.006	*	*	*	0.001	0.031
	Total Degree	0.134	0.131	0.04	0.069	0.083	0.121	0.859

113	Betweenness Centrality	0.005	0.008	0	*	0.002	0.006	0.053
	Eigenvector Centrality	0.018	0.048	0	*	*	0.005	0.268
	In-Degree	0.061	0.119	0	0.005	0.016	0.047	0.831
	Local Clustering Coefficient	0.527	0.151	0.118	0.443	0.553	0.64	0.837
	Out-Degree	0.061	0.009	0.031	0.055	0.062	0.068	0.083
	PageRank	0.003	0.005	*	*	*	0.001	0.032
	Total Degree	0.122	0.12	0.036	0.065	0.078	0.107	0.891
114	Betweenness Centrality	0.005	0.008	0	*	0.002	0.006	0.048
	Eigenvector Centrality	0.02	0.048	0	*	*	0.008	0.259
	In-Degree	0.065	0.122	0	0.005	0.019	0.057	0.708
	Local Clustering Coefficient	0.534	0.144	0.143	0.469	0.558	0.633	0.827
	Out-Degree	0.065	0.01	0.03	0.06	0.065	0.073	0.092
	PageRank	0.003	0.005	*	*	*	0.002	0.029
	Total Degree	0.13	0.124	0.043	0.07	0.086	0.123	0.765
115	Betweenness Centrality	0.005	0.008	0	*	0.002	0.007	0.057
	Eigenvector Centrality	0.019	0.048	0	*	*	0.009	0.262
	In-Degree	0.064	0.12	0	0.005	0.016	0.056	0.764
	Local Clustering Coefficient	0.53	0.147	0.138	0.451	0.558	0.64	0.833
	Out-Degree	0.064	0.01	0.024	0.056	0.064	0.072	0.091
	PageRank	0.003	0.005	*	*	*	0.002	0.031
	Total Degree	0.128	0.122	0.038	0.067	0.083	0.125	0.828

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